

Methods

Decision Making with Side Information: A Causal Transport Robust Approach

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Abstract. We consider stochastic optimization with side information where, prior to decision making, covariate data are available to inform better decisions. To hedge against data uncertainty while capturing the information structure revealed from the conditional distribution of random problem parameters given the covariate values, we propose a distributionally robust formulation based on causal transport distance. We derive a dual reformulation for evaluating the worst-case expected cost and show that the worst-case distribution in a causal transport distance ball preserves the conditional information structure from the nominal distribution. When optimizing over affine decision rules, we identify cases in which the overall problem can be solved via convex programming. When optimizing over all (non-parametric) decision rules, we identify a new class of robust optimal decision rules when the cost function is convex with respect to a one-dimensional decision variable.

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1. Introduction

Stochastic optimization with side information, also known as contextual stochastic optimization or conditional stochastic optimization, addresses the following problem:

$$\min_{w \in \mathcal{D}} \mathbb{E}[\Psi(w, Z) | X = x], \quad (1)$$

where the goal is to select a decision w from a feasible set $\mathcal{D}$ that minimizes the conditional expectation of the cost $\Psi(w, Z)$, dependent on both the decision w and a random variable Z , given some side information, represented by a covariate X . The increasing use of side information from covariate data has significantly enhanced decision making across e-commerce and online platforms, enabling more personalized informed strategies. The performance evaluation often encompasses the entire covariate population; for example, the manager of an e-commerce company cares about overall performance across all customer types. As such, we

are interested in finding a decision rule that minimizes the expected cost over the joint distribution of the covariate X and the random variable Z :

$$\min_{f \in \mathcal{F}} \mathbb{E}[\Psi(f(X), Z)]. \quad (2)$$

The decision rule f offers an end-to-end map from the covariate space $\mathcal{X}$ to the decision space $\mathcal{D}$, chosen from a family $\mathcal{F}$ of functions—parametric or nonparametric—on $\mathcal{X}$. Such an end-to-end approach is often computationally efficient at deployment time, because the decision for a new context x can be obtained by directly evaluating $f(x)$. Moreover, the complexity of the policy class $\mathcal{F}$ can be adjusted flexibly—ranging from simple parametric forms such as affine decision rules to rich nonparametric families—allowing practitioners to strike a desired balance between interpretability, flexibility, and statistical efficiency.

Formulation (2) covers many contextual optimization problems in operations research and machine

learning. For instance, suppose $\Psi(w, z) = h(w - z)_+ + b(z - w)_+$, where w is the order quantity decision, z represents the demand of a product, $(\cdot)_+ = \max(\cdot, 0)$, and $h, b \geq 0$ represent the overage cost and the underage cost, respectively, then (2) is known as the big-data newsvendor model (Ban and Rudin 2019). If $\mathcal{F}$ is the set of all measurable functions on $\mathcal{X}$, then the optimal order quantity equals the conditional critical fractile $f^*(x) = F_x^{-1}(b/(h+b))$, where F_x is the conditional cumulative distribution function of demand Z given $X = x$; and if $\mathcal{F}$ is the set of affine functions on $\mathcal{X}$, then (2) finds the optimal affine decision rule for the big-data newsvendor. As another example, when $\Psi(w, z) = (w - z)^2$ and $\mathcal{F}$ is the set of all measurable functions on $\mathcal{X}$, the optimal solution to (2) is $f^*(x) = \mathbb{E}[Z|X = x]$, and thus Formulation (2) finds the conditional mean of Z given X . More examples will be given in Section 2.2. We note that this is not the only formulation of contextual decision making, and we discuss related work in Section 1.3.

Similar to classical stochastic optimization, the underlying joint distribution $\mathbb{P}_{\text{true}}$ of (X, Z) is often not known exactly; instead, historical data from the distribution are available. In this case, a natural approach is to replace the unknown underlying distribution of demands and features with their empirical counterpart. However, if the function class $\mathcal{F}$ contains all measurable functions, then the resulting empirical risk minimization problem yields a degenerate decision rule that is defined only on the set of historical observations of the contextual values. To overcome the degeneracy issue, it is reasonable to consider a data-driven, distributionally robust contextual decision-making framework

$$\min_{f \in \mathcal{F}} \max_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}}[\Psi(f(X), Z)], \quad (3)$$

a minimax formulation that hedges the data uncertainty, where $\mathfrak{M}$ is a chosen uncertainty set. At the core of the distributionally robust formulation is the choice of the uncertainty set, and the presence of the side information adds new challenges beyond those for classic stochastic optimization. In Section 1.1, we review some existing choices of uncertainty sets and discuss their potential issues.

1.1. Discussion on Some Existing Uncertainty Sets

To begin with, we would like to focus on distance-based uncertainty sets, as the other popular choice—moment-based uncertainty sets—generally lacks statistical consistency.

Two classes of distance-based uncertainty sets have been widely studied in the literature. The first class is the divergence family, deeply rooted in statistics, information theory, and physics. Consider the following example.

Example 1 (Kullback-Leibler Robust Solution is Degenerate). Suppose $\mathfrak{M}$ is a Kullback-Leibler (KL) divergence ball, centered at the empirical distribution $\hat{\mathbb{P}}$ constructed from K independently and identically distributed (i.i.d.) samples from a continuous underlying distribution. Then with probability of one, $\hat{\mathbb{P}}$ can be represented as $1/K \sum_{k=1}^K \delta_{(\hat{x}_k, \hat{z}_k)}$, where K is the sample size and all $(\hat{x}_k, \hat{z}_k)$ s are different from each other. Let $\mathcal{F}$ be the set of all measurable functions on $\mathcal{X}$. Then, we claim that the KL robust optimal solution would satisfy

$$f_{\text{kl}}(x) = \begin{cases} \arg \min_{w \in \mathcal{D}} \Psi(w, \hat{z}_k), & \text{if } x = \hat{x}_k, k = 1, \dots, K, \\ \text{arbitrary value}, & \text{otherwise.} \end{cases}$$

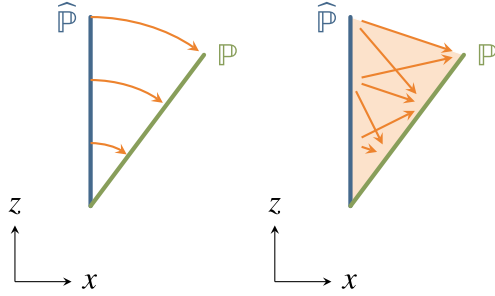
Indeed, every distribution in the KL ball is supported only on the data points from $\hat{\mathbb{P}}$ but may differ from it in the probability weights. On an in-sample data point $\hat{x}_k$, regardless of its weight, the optimal decision would always be the minimizer of $\Psi(\cdot, \hat{z}_k)$ due to the interchangeability principle (Shapiro et al. 2014, section 9.3.4). Furthermore, because the KL robust cost depends only on the function values on the in-sample data, the robust optimal solution can take any value on out-of-sample data without changing the objective value.

Example 1 shows that the KL robust optimal decision rule is degenerate with probability of one when the underlying distribution is continuous, regardless of the size of the uncertainty set, the sample size, or the objective function. A similar phenomenon also holds for all other divergence measures due to the structure of the worst-case distribution (Bayraksan and Love 2015).

The second class is the Wasserstein, or transport cost distance, family. It is well known that the resulting uncertainty set avoids some degeneracy issues of the divergence sets in stochastic optimization (Kuhn et al. 2019, Gao and Kleywegt 2023). Nonetheless, it faces new challenges when additional side information is presented. Let us consider the following toy example.

Example 2 (Wasserstein Set Cannot Capture Conditional Information). In Figure 1, $\hat{\mathbb{P}}$ and $\mathbb{P}$ are two uniform distributions supported, respectively, on the vertical and rotated line segments with a common endpoint with x -entry $\hat{x}$. The angle between the two line segments is ε radian. Notably, the conditional distribution $\mathbb{P}_{Z|X=\hat{x}}$ is a Dirac measure for $x > \hat{x}$, which is apparently very different from the conditional distribution $\hat{\mathbb{P}}_{Z|X=\hat{x}}$ that is uniform on the blue line segment. As will be calculated in Section 2, the Wasserstein distance between $\hat{\mathbb{P}}$ and $\mathbb{P}$ is $O(\varepsilon)$, and the optimal transport map is a rotation. This means a Wasserstein ball centered at $\hat{\mathbb{P}}$ would always contain a distribution that has a different conditional information structure than that of $\hat{\mathbb{P}}$, regardless

Figure 1. (Color online) $\hat{\mathbb{P}}$ and $\mathbb{P}$ Have Completely Different Conditional Information Structures but with $O(\varepsilon)$ Wasserstein Distance



Note. If we restrict transport plans to causal ones, then two distributions are distinguished by an $O(1)$ distance.

of the value of ε . On the other hand, as will be revisited in Section 2.1, by restricting to the causal transport map (shown in the right plot) which, in this case, is the independent joint distribution $\hat{\mathbb{P}} \otimes \mathbb{P}$, distributions with a different conditional information structure will be ruled out from the uncertainty set.

In practice, the following situation is often observed in the data: The conditional distribution can be estimated accurately for a number of covariate values but is largely unobserved for other values. For example, historical data may provide a good estimate of the conditional demand distribution for the product sold at deployed vending machines, but demand at new locations remains unexplored. Nonetheless, it is conceivable that the conditional demand distribution should resemble that of similar locations. In such cases, it would be reasonable to expect that the conditional distributions $\mathbb{P}_{Z|X=x}$ and $\mathbb{P}_{Z|X=\hat{x}}$ corresponding to two similar values x and $\hat{x}$ should be close in a certain way. Therefore, we would like to choose an uncertainty set containing distributions with a similar conditional information structure to the nominal distribution. Example 2 demonstrates that the Wasserstein uncertainty set fails to preserve the conditional information structure and, in fact, the same phenomenon also holds for the worst-case distribution, as will be shown in Section 3.2. This raises concerns about the conservatism of the Wasserstein formulation.

1.2. Our Contributions

To capture the conditional information, in this paper, we consider a distributional uncertainty set based on *causal transport distance*, a notion that is related to Wasserstein distance but imposes an additional assumption on the transport plan; see Section 2.1 for its definition and a more in-depth discussion. The causal transport distance uncertainty set brings new computational challenges to the inner optimization over probability distributions in (3), which require new analysis of

tractable reformulations and interpretations. Moreover, when the outer minimization over the class of decision rules is performed over a nonparametric class, additional computational challenges arise due to the associated infinite-dimensional functional optimization. Our main contributions are as follows.

(I) We develop a strong duality reformulation for computing the worst-case loss of a fixed decision rule (Section 3.1). Our proof is based on a new analysis of the worst-case distribution, which demonstrates how our choice of distributional uncertainty set captures the conditional information structure of the random variable given the side information (Section 3.2).

(II) We study tractable reformulations for finding the optimal decision rule when optimizing over (i) the affine class, (ii) reproducing kernel Hilbert space, and (iii) all (nonparametric) decision rules. We provide convex reformulations when $\Psi(w, z)$ is convex in the decision variable w , in particular for (i) when Ψ is linear in w or bilinear/quadratic in w and z (Section 4.1), for (ii) when Ψ is Lipschitz in w (Section 4.2), and for (iii) when the decision w is one dimensional. This provides a new class of decision rule with no suboptimality gap for adjustable robust optimization (Section 4.3). We illustrate our results with conditional mean estimation, feature-based newsvendor, personalized pricing, and contextual linear optimization.

(III) We conduct numerical experiments to demonstrate that the causal transport distance uncertainty set effectively utilizes conditional information compared with the Wasserstein uncertainty set and compare the performance of different classes of decision rules (Section 5).

1.3. Related Literature

1.3.1. On Stochastic Optimization with Side Information. In the literature, the frameworks for contextual optimization (with an offline data set) can be broadly classified into three categories (Qi and Shen 2022, Sadana et al. 2025): *sequential learning and optimization*, *integrated prediction and optimization*, and *decision rule optimization*.

(I) Sequential learning and optimization (or two-step optimization) involves first estimating the conditional distribution of Z given a new context $X = x$ and then optimizing for the conditional expectation $\min_{w \in \mathcal{D}} \mathbb{E}[\Psi(w, Z) | X = x]$ (Toktay and Wein 2001, Zhu and Thonemann 2004). This approach has theoretical guarantees discussed in El Balghiti et al. (2019) and Hu et al. (2022). Recent advancements in this area include approaches using Dirichlet processes (Hannah et al. 2010), Nadaraya-Watson kernel regression (Hanasusanto and Kuhn 2013, Ban and Rudin 2019, Wang et al. 2026), local regression and classification (Bertsimas and McCord 2019, Bertsimas and Kallus 2020), trees and forests (Ban et al. 2019, Bertsimas and McCord 2019, Bertsimas and Kallus 2020), and empirical residuals (Kannan et al. 2024,

2025). Additionally, robustness in optimization and regularization have been explored in Tulabandhula and Rudin (2013), Zhu et al. (2022), Bertsimas and Van Parys (2022), Loke et al. (2020), Esteban-Pérez and Morales (2022), Van Parys and Bennouna (2022), Van Parys et al. (2021), Chenreddy et al. (2022), Nguyen et al. (2025), and Perakis et al. (2023). However, as noted in Liyanage and Shanthikumar (2005) and Ban and Rudin (2019), statistical estimation errors and model misspecifications can propagate through the optimization process, resulting in suboptimal performance, whereas in the case of contextual linear optimization, Hu et al. (2022) show that the plug-in approach actually achieves faster regret convergence rates than methods that directly optimize downstream decision performance.

(II) Integrated prediction and optimization avoid estimating the conditional distribution by optimizing the parameterization of a contextual predictor to minimize downstream expected costs $\mathbb{E}[\Psi(w, Z) | X = x]$. Various estimation techniques are explored here, including models based on conditional distributions (Kallus and Mao 2023, Qi et al. 2025), regret minimization (Estes 2021), smart prediction-then-optimization (El Balghiti et al. 2019, Elmachtoub et al. 2020, Elmachtoub and Grigas 2022, Ho-Nguyen and Kılınç-Karzan 2022), bilevel optimization (Cao and Gao 2021, Muñoz et al. 2022, Hu et al. 2024), inverse optimization, and so on. This approach requires solving an optimization over the decisions for each individual context.

(III) Decision rule optimization is an end-to-end formulation that finds a decision rule prescribing the decision for every possible context. Because of the computational difficulty of this infinite-dimensional optimization, typically the policies are parameterized by a finite-dimensional vector, such as coefficients in an affine function of features (Brandt et al. 2009, Ban and Rudin 2019, Bazier-Matte and Delage 2020, Bertsimas et al. 2023) or in a reproducing kernel Hilbert space (Bertsimas and Koduri 2022) and weight matrices in a neural network (Oroojlooyjadid et al. 2020, Liu et al. 2021, Qi et al. 2023, Rychener et al. 2023).

Our formulation falls into the third category, but our results in Section 4 do not necessarily restrict the class of decision rules to a parametric family. In this respect, the closest work to ours is Zhang et al. (2024), which considers robust optimization over decision rules with the Wasserstein uncertainty set; see the last paragraph of the literature review for a detailed comparison. We remark that in the online setting, stochastic optimization with side information has also been considered under the umbrella of contextual bandits and reinforcement learning, and there have been some studies of decision-dependent uncertainty (Basciftci et al. 2021, Yu and Shen 2022, Vayanos et al. 2025). These are beyond the scope of this paper.

1.3.2. On Transport-Distance Based Distributionally Robust Optimization. Distributionally robust optimization (DRO) has received significant attention recently as a tool for decision-making under uncertainty, and different approaches mainly differ in how the uncertainty set is constructed. We refer to Rahimian et al. (2019) for a thorough review of choices of uncertainty set. Our choice of uncertainty set is aligned with DRO with transport distance, such as Wasserstein distance (Pflug and Wozabal 2007, Wozabal 2012, Esfahani and Kuhn 2018, Blanchet and Murthy 2019, Blanchet et al. 2019, Gao 2023, Gao and Kleywegt 2023, Gao et al. 2024b) and nested distance (Rüschendorf 1985, Analui and Pflug 2014, Pichler and Shapiro 2021)—a symmetrized analogue of causal transport distance. The origin of causal transport could be traced back to the Yamada-Watanabe criterion for stochastic differential equations (Yamada and Watanabe 1971, Jean 1980, Kurtz 2014). In optimal transport theory, Lassalle (2018) investigated the transport problem in continuous time under the causal constraints, and Backhoff et al. (2017) studied a discrete-time analogue. Causal transport has been applied to continuous-time stochastic optimization in Acciaio et al. (2020), as well as other areas such as stochastic control (Acciaio et al. 2019) and machine learning (Xu et al. 2020). In discrete time stochastic programming, the nested distance has been exploited to study the stability and sensitivity of multistage stochastic programming (Pflug 2010; Pflug and Pichler 2012, 2014, 2015, 2016; Bartl and Wiesel 2023).

Our problem can be viewed as a two-stage DRO with causal transport distance. After our paper's first draft appeared online, several works studied DRO with causal transport distance. Gao et al. (2024a) studied the dynamic programming reformulation for multistage DRO with nested distance. Jiang (2024) derives duality for the DRO problem with causal transport penalty. Compared with their methodology, our constructive proof of duality enables the characterization of the worst-case distribution, and we develop tractable reformulations for decision-rule optimization.

Another related work is Shen et al. (2024), which proposes a 2-Wasserstein distributionally robust contextual bandit framework that places a Wasserstein ball around the context distribution together with a family of Wasserstein balls for the conditional reward distributions. From this perspective, our dual reformulation is closely related. However, their ambiguity sets impose hard constraints with a shared radius across all contexts, leading to a two-level nested optimization.

1.3.3. On Decision-Rule Approach in Adjustable Robust Optimization. In the literature for adjustable robust optimization, different choices of decision rules have been thoroughly investigated, including affine families (Chen et al. 2008; Bertsimas et al. 2010, 2011,

2022; Bertsimas and Goyal 2012; Iancu et al. 2013; El Housni and Goyal 2021; Georghiou et al. 2025), k -adaptability (Hanasusanto et al. 2015, 2016; Subramanyam et al. 2019), iterative splitting of uncertainty sets (Postek and Hertog 2016), binary decision rules (Bertsimas and Georghiou 2015), nonparametric Markovian stopping rules (Sturt 2023), and so on. Most of these works do not consider side information in their problem formulations. Bertsimas et al. (2023) consider dynamic decision making with side information using affine decision rules, whereas we consider general decision rules in a static setting, and Zhang et al. (2024) consider the newsvendor problem with Wasserstein distance, whereas we consider a different uncertainty set, and we adopt a completely different proof strategy and obtain a broader class of optimal policies for adjustable robust optimization that encapsulates the Shapley policy proposed therein.

The rest of the paper proceeds as follows. We introduce the causal transport distance and corresponding robust model in Section 2. In Section 3, we develop a duality result for evaluating the worst-case expected cost by exploiting the structure of the worst-case distribution. In Section 4, we consider the outer optimization over affine decision rules and over all decision rules. Finally, we present numerical results in Section 5 and conclude the paper in Section 6. Proofs and additional results are deferred to the online appendices.

2. Distributionally Robust Optimization with Causal Transport Distance

In this section, we briefly introduce notation and provide some background on distributionally robust optimization with causal transport distance.

Notation. Let $(\mathcal{X}, \|\cdot\|_{\mathcal{X}})$, $(\mathcal{Z}, \|\cdot\|_{\mathcal{Z}})$ be subsets of normed vector spaces. For notational simplicity, the subscripts in $\|\cdot\|_{\mathcal{X}}$ and $\|\cdot\|_{\mathcal{Z}}$ will be omitted as long as they can be inferred from the context. Let $p \in [1, \infty)$ and denote by q its Hölder conjugate number, that is, $1/p + 1/q = 1$. We denote by $\mathcal{P}_p(\mathcal{Z})$ the set of probability measures of $\mathcal{Z}$ with finite p th moment, namely, $\mathbb{Q} \in \mathcal{P}_p(\mathcal{Z})$ if and only if $\mathbb{E}_{Z \sim \mathbb{Q}}[\|Z\|^p] < \infty$. The support of a distribution is denoted by $\text{supp } \mathbb{Q}$. The set of all possible transport plans between the given marginals $\mathbb{Q}_1, \mathbb{Q}_2 \in \mathcal{P}(\mathcal{X} \times \mathcal{Z})$, on the product space $(\mathcal{X} \times \mathcal{Z}) \times (\mathcal{X} \times \mathcal{Z})$, is denoted as $\Gamma(\mathbb{Q}_1, \mathbb{Q}_2)$.

2.1. Causal Transport Distance

Our motivation to adopt the causal transport distance in DRO is illustrated by the following example. Consider the feature-based newsvendor problem, where historical demand for a product in a vending machine is influenced by covariates such as location, weather, and economic conditions. In such a scenario, the causal relationship is directed: The distributional uncertainty

of the features can lead to uncertainty in the demand but not vice versa. Therefore, if we consider a data perturbation map

$$T : (\hat{X}, \hat{Z}) \mapsto T(\hat{X}, \hat{Z}) = (T_1(\hat{X}, \hat{Z}), T_2(\hat{X}, \hat{Z})),$$

the perturbation of features (e.g., location, weather, economic state) should not depend on the demand, but the perturbation of demand can be affected by the perturbation of features. In other words, the perturbation map should have the form

$$T(\hat{X}, \hat{Z}) = (T_1(\hat{X}), T_2(\hat{X}, \hat{Z})),$$

where $\hat{X}$ is transported to $T_1(\hat{X})$, and given $\hat{X}$, $X = T_1(\hat{X})$ is a constant. This implies that X is conditionally independent of $\hat{Z}$, represented as $X \perp \hat{Z} | \hat{X}$. Extending upon this notion of conditional independence, we introduce the following definition of causal transport plan and causal transport distance.

Definition 1 (Causal Transport Distance). A joint distribution $\gamma \in \Gamma(\hat{\mathbb{P}}, \mathbb{P})$ is called a *causal transport plan* if, for $((\hat{X}, \hat{Z}), (X, Z)) \sim \gamma$, X and $\hat{Z}$ are conditionally independent given $\hat{X}$:

$$X \perp \hat{Z} | \hat{X}.$$

We denote by $\Gamma_c(\hat{\mathbb{P}}, \mathbb{P})$ the set of all transport plans $\gamma \in \Gamma(\hat{\mathbb{P}}, \mathbb{P})$ that are causal. Let $p \in [1, \infty)$. The p -causal transport distance between $\hat{\mathbb{P}}$ and $\mathbb{P}$ is defined as

$$C_p(\hat{\mathbb{P}}, \mathbb{P}) := \left(\inf_{\gamma \in \Gamma_c(\hat{\mathbb{P}}, \mathbb{P})} \mathbb{E}_{((X, Z), (\hat{X}, \hat{Z})) \sim \gamma} [\|X - \hat{X}\|^p + \|Z - \hat{Z}\|^p] \right)^{1/p}.$$

The conditional independence condition in Definition 1 basically means that the destination X of a sample in a causal transport plan should depend only on the origin $\hat{X}$ but not on the associated information of $\hat{Z}$. There are other equivalent definitions of a causal transport plan, which are provided in Online Appendix EC.1.

Like the Wasserstein distance, the causal transport distance finds the minimal transport cost between two distributions, where norms capture the geometry of the data space and the similarity between samples. Nevertheless, causal transport distance differs from Wasserstein distance in the involved class of transport plans: Wasserstein distance considers all transport plans with given marginals, whereas causal transport distance restricts causal transport plans as defined in Definition 1.

Let us use the following example to visually explain a causal transport plan.

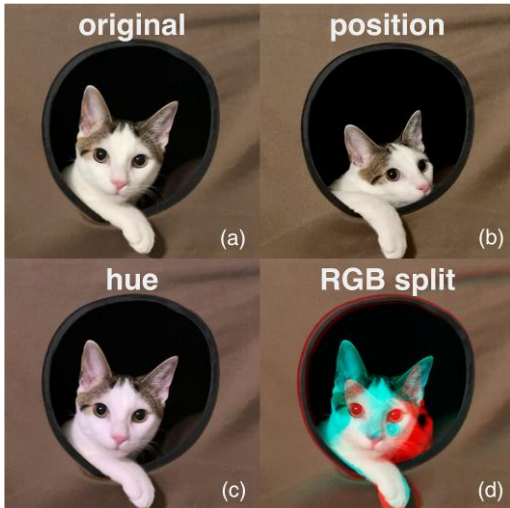
Example 3 (Causal Transport Between Colored Images). Let $\mathcal{X} = \{1, 2, \dots, H\}^2$, where H represents the width of a squared image, and let $\mathcal{Z} = \{R, G, B\}$, representing the three color channels, red (R), green (G), and blue (B). One can identify $\mathcal{Z}$ with the three-basis vector of $\mathbb{R}^3$. A bitmap

image stores the position-color information of an image via an $H \times H \times 3$ tensor $A = (A_{ijk})_{i,j \in \{1,2,\dots,H\}, k \in \{1,2,3\}}$. Its (i,j,k) th entry $A_{ijk} \in \{0,1,\dots,255\}$ represents the eight-bit indexed color at pixel position (i,j) in the k th channel. With a normalizing constant $M = \sum_{i,j,k} A_{ijk}$, the tensor A/M represents a probability mass function on $\mathcal{X} \times \mathcal{Z}$. Let us equip norms $\|\cdot\|_{\mathcal{X}} = \|\cdot\|_{\ell^1}$ and $\|\cdot\|_{\mathcal{Z}} = cH\|\cdot\|_{\ell^\infty}$, where c is a scaling parameter. Figure 2 contains four images of a cat: (a), (b), and (c) can be viewed as real natural images with different poses or lighting conditions, whereas (d) can be viewed as an artificial image in which the pose exhibited via the red channel is different from that via the green/blue channel.

(I) The movement of the cat yields a causal transport plan from (a) to (b), as under such movement, the destination (X,Z) in (b) of a position-channel pair $(\hat{X}, \hat{Z})$ in (a) depends only on its original position $\hat{X}$ but not on the channel information $\hat{Z}$, or put it differently, all channels are moved in the same way from $\hat{X}$ to X without changing the channel value $\hat{Z}$. This matches precisely the definition of causal transport.

(II) The cats in (a) and (c) have identical poses but different hue values. Changing the hue values of an image would affect its RGB values and, in turn, the distribution on $\mathcal{Z}$. Such color adjustment (changing RGB values while fixing the position) defines a causal transport plan from (a) to (c). Indeed, under such a movement, a position-channel pair $(\hat{X}, \hat{Z})$ in (a) keeps its position in $\mathcal{X}$, namely, $X = \hat{X}$, regardless of the value of $\hat{Z}$. Note that in a causal transport plan, we allow the destination Z of $\hat{Z}$ to depend on both $\hat{X}$ and $\hat{Z}$, that is, at each position in the image, changes in color are permitted.

Figure 2. (Color online) Cat Images with Different Poses or Lighting Conditions



Note. An image (a) and its variations by shifting the position (b), adjusting the hue (c), or splitting the RGB channels (d).

(III) The green and blue channels of (d) have the same pose as (a), whereas the red channel of (d) has the same pose as (b). If we consider a transport plan that keeps a position-channel pair $(\hat{X}, \hat{Z})$ if $\hat{Z} \in \{G, B\}$, and transport it according to the cat's movement if $\hat{Z} = R$, then such a transport plan is *not causal*, because given $\hat{X}$, where this position-channel pair is transported, depends on the channel information $\hat{Z}$.

In Table 1, we compute the Wasserstein distance and causal transport distance between Figure 2(a) and the other three variations, with $H = 32$ and $c = 4$. We find that the causal transport distance between Figure 2(a) and the artificial image Figure 2(d) is much larger than that between Figure 2(a) and natural images Figure 2, (b) and (c). In contrast, the Wasserstein distance fails to capture such an intuition.

As hinted in Example 3, one of the main advantages of causal transport distance over Wasserstein distance is that it captures the structure of the conditional distribution. To further illustrate this, let us revisit the toy Example 2.

Example 2 (Revisited). We compute the causal transport distance and the Wasserstein distance between $\hat{\mathbb{P}}$ and $\mathbb{P}$ shown in Example 2. Because the conditional distribution of $\mathbb{P}$ is a Dirac measure for every x , the causal transport distance between $\hat{\mathbb{P}}$ and $\mathbb{P}$ is uniformly bounded from below by a positive constant for all $\varepsilon > 0$. In fact, it is not hard to see that the only causal transport plan is the independent joint distribution $\hat{\mathbb{P}} \otimes \mathbb{P}$, so

$$\begin{aligned} C_p(\hat{\mathbb{P}}, \mathbb{P})^p &= \frac{1}{\sin \varepsilon} \int_0^{\sin \varepsilon} |x - 0|^p dx \\ &\quad + \frac{1}{\cos \varepsilon} \int_0^1 \int_0^{\cos \varepsilon} |\hat{z} - z|^p dz d\hat{z} \\ &= \frac{\sin^p \varepsilon}{p+1} + \frac{1 + \cos^{p+2} \varepsilon - (1 - \cos \varepsilon)^{p+2}}{(p+1)(p+2)\cos \varepsilon} \\ &= \left((1+p)\left(1 + \frac{p}{2}\right) \right)^{-\frac{1}{p}} + O(\varepsilon). \end{aligned}$$

As a result, $\mathbb{P}$ would not belong to the uncertainty set induced by the causal transport distance with a small radius. This is consistent with our intuition. In contrast, for the Wasserstein distance, observe that the optimal transport plan is simply the rotation transform; thereby, the

Table 1. Distance Between Figure 2(a) and the Other Three Variations

Variations	(b)	(c)	(d)
Wasserstein distance	2.434	2.109	0.677
Causal transport distance	2.543	2.161	7.489

Wasserstein distance is $(p+1)^{-1/p}(\sin^p \varepsilon + (1 - \cos \varepsilon)^p)^{1/p} = O(\varepsilon)$, which is small whenever the angle between the two line segments is small. Consequently, any Wasserstein uncertainty set with a positive radius contains infinitely many distributions with dramatically different conditional information structures from the nominal one and therefore may lead to an overly conservative solution.

Next, we point out an important property of the uncertainty set constructed using the causal transport distance: For any $\hat{\mathbb{P}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z})$ and $\rho > 0$, the set $\mathfrak{M} = \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z}) : C_p(\hat{\mathbb{P}}, \mathbb{P}) \leq \rho\}$ is convex, as indicated in the following lemma.

Lemma 1 (Convexity). *If $\gamma^{(0)}$ and $\gamma^{(1)}$ are two causal transport plans from $\hat{\mathbb{P}}$ to $\mathbb{P}^{(0)}$ and $\mathbb{P}^{(1)}$, respectively, then for any $q \in [0, 1]$, $\gamma^{(q)} := (1 - q)\gamma^{(0)} + q\gamma^{(1)}$ is also a causal transport plan from $\hat{\mathbb{P}}$ to $\mathbb{P}^{(q)} = (1 - q)\mathbb{P}^{(0)} + q\mathbb{P}^{(1)}$. Moreover, everything follows even if we replace q by any measurable function $q : \mathcal{X} \rightarrow [0, 1]$.*

We remark that the direction of the transport plan matters: If $\gamma^{(0)}$ and $\gamma^{(1)}$ are two causal transport plans from $\hat{\mathbb{P}}^{(0)}$ and $\hat{\mathbb{P}}^{(1)}$ to $\mathbb{P}$, respectively, we cannot assert that their convex combination $\gamma^{(q)}$ is also a causal transport plan. For a counterexample, please refer to figure 1.17 in Pflug and Pichler (2014).

2.2. Distributionally Robust Formulation

Based on the definition in the previous section, we study the following distributionally robust optimization problem with causal transport distance

$$v_P := \inf_{f \in \mathcal{F}} \max_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}}[\Psi(f(X), Z)],$$

where $\mathfrak{M} = \{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z}) : C_p(\hat{\mathbb{P}}, \mathbb{P}) \leq \rho\}$. (P)

Below, we list a few examples.

Example 4 (Conditional Mean Estimation). The conditional mean of Z given X can be estimated by minimizing the square loss $(f(X) - Z)^2$. Thus, we consider the following robust conditional mean estimation problem:

$$\inf_{f \in \mathcal{F}} \sup_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}}[(f(X) - Z)^2].$$

Example 5 (Feature-Based Newsvendor). Let h and b represent the unit overage cost and the unit underage cost, respectively, and let Z be the random demand and X be the covariate features. The goal is to minimize the newsvendor cost function $\Psi(w, z) = h(w - z)_+ + b(z - w)_+$. Consider

$$\inf_{f \in \mathcal{F}} \sup_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}}[h(f(X) - Z)_+ + b(Z - f(X))_+].$$

Note that this model also serves as the conditional $\frac{b}{b+h}$ -quantile estimation. In particular, when $h = b = 1$, this is the conditional median estimation.

Example 6 (Personalized Pricing). Consider an affine demand model $D(w) = z_1 w + z_2 = Z^\top \begin{pmatrix} w \\ 1 \end{pmatrix}$, where w is the price and z are unknown coefficients, with $z_2 > 0$ representing the demand at zero price and $z_1 < 0$ representing the price sensitivity coefficient, which is the rate at which the price affects the demand. In practice, both coefficients z_1 and z_2 may exhibit heterogeneity across populations. As such, we model it as a two-dimensional random variable Z that is affected by the contextual information X , which the decision maker can use to adjust the price directly or indirectly through personalized promotion. The revenue is calculated as $w(Z_1 w + Z_2)$. Consider revenue maximization with personalized pricing

$$\inf_{f \in \mathcal{F}} \sup_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}} \left[-f(X) Z^\top \begin{pmatrix} f(X) \\ 1 \end{pmatrix} \right].$$

In the last example, we consider a contextual linear optimization problem where the decision rule is restricted to be affine.

Example 7 (Contextual Linear Optimization with Affine Decision Rule). Consider a contextual linear optimization problem in which one minimizes the loss function $\Psi(w, z) = w^\top z$. Take a linear policy class $\mathcal{F}_\Theta$ defined by

$$\mathcal{F}_\Theta = \{x \mapsto B^\top x + \delta : (B, \delta) \in \Theta\},$$

where $\Theta = \{(B, \delta) \in \mathbb{R}^{d \times m} \times \mathbb{R}^m : B^\top x + \delta \in \mathcal{D}, \forall x \in \mathcal{X}\}$,

(4)

so that $f(\mathcal{X}) \subset \mathcal{D}$ for each $f \in \mathcal{F}_\Theta$. The robust contextual linear optimization problem is given by

$$\inf_{f \in \mathcal{F}_\Theta} \sup_{\mathbb{P} \in \mathfrak{M}} \mathbb{E}_{(X, Z) \sim \mathbb{P}}[f(X)^\top Z].$$

3. Evaluating the Worst-Case Expectation

In this section, we develop a tractable reformulation for the inner maximization of (P) based on strong duality. As a by-product of our proof, we also derive the structure of the worst-case distribution, demonstrating that our choice of a distributional uncertainty set based on causal transport distance helps preserve the conditional information structure of the nominal distribution in the worst case.

Throughout this paper, we make the following assumption, which focuses on the data-driven setting where the nominal distribution is discrete, although our proof technique can be extended to a general metric space with additional technical treatment.

Assumption 1. Assume $\mathcal{X}$, $\mathcal{Z}$, and $\mathcal{D}$ are subsets of normed vector spaces. The cost function $\Psi : \mathcal{D} \times \mathcal{Z} \rightarrow \mathbb{R}$ is measurable. The nominal distribution $\hat{\mathbb{P}} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z})$ is a discrete probability measure

$$\hat{\mathbb{P}} = \sum_{k=1}^K \sum_{i=1}^{n_k} \hat{p}_{ki} \delta_{(\hat{x}_{ki}, \hat{z}_{ki})}, \quad \text{with } \sum_{k=1}^K \sum_{i=1}^{n_k} \hat{p}_{ki} = 1.$$

3.1. Strong Duality Reformulation

We begin by developing a tractable reformulation by deriving its strong dual. For a fixed decision rule f , we define the primal problem as

$$v_P^f := \max_{\mathbb{P} \in \mathcal{M}} \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)], \quad (P^f)$$

and the dual problem as

$$v_D^f := \inf_{\lambda \geq 0} \left\{ \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} \left[\sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda \|z - \hat{Z}\|^p \} \right] \right\} \right] \right. \\ \left. - \lambda \|x - \hat{X}\|^p \right\}. \quad (D^f)$$

The dual variable λ corresponds to the Lagrangian multiplier of the causal constraint in the primal problem. We will show that (P^f) and (D^f) are equal, leading to the main result of Theorem 1 by taking the infimum over f .

To prove the strong duality, we first develop a relatively straightforward weak duality result.

Proposition 1 (Weak Duality). *Let $f : \mathcal{X} \rightarrow \mathcal{D}$ be a measurable function. Then $v_P^f \leq v_D^f$.*

Proof. The proof is based on an application of Lagrangian weak duality. First, we derive from the Lagrangian weak duality the following:

$$\begin{aligned} v_P^f &= \sup_{\mathbb{P}} \{ \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] : C_p(\hat{\mathbb{P}}, \mathbb{P})^p \leq \rho^p \} \\ &= \sup_{\mathbb{P}} \inf_{\lambda \geq 0} \{ \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] - \lambda (C_p(\hat{\mathbb{P}}, \mathbb{P})^p - \rho^p) \} \\ &\leq \inf_{\lambda \geq 0} \sup_{\mathbb{P}} \{ \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] - \lambda (C_p(\hat{\mathbb{P}}, \mathbb{P})^p - \rho^p) \}. \end{aligned}$$

Because for any $\gamma \in \Gamma_c(\hat{\mathbb{P}}, \mathbb{P})$,

$$\mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] = \mathbb{E}_{((X,Z), (\hat{X}, \hat{Z})) \sim \gamma} [\Psi(f(X), Z)],$$

so we can write

$$\begin{aligned} &\mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] - \lambda (C_p(\hat{\mathbb{P}}, \mathbb{P})^p - \rho^p) \\ &= \lambda \rho^p + \sup_{\gamma \in \Gamma_c(\hat{\mathbb{P}}, \mathbb{P})} \mathbb{E}_{\gamma} [\Psi(f(X), Z) - \lambda \|X - \hat{X}\|^p - \lambda \|Z - \hat{Z}\|^p]. \end{aligned}$$

By the tower property,

$$\begin{aligned} \mathbb{E}_{\gamma}[\cdot] &= \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} [\mathbb{E}_{\gamma_{X|\hat{X}}} [\mathbb{E}_{\gamma_{Z|\hat{X}, X}} [\mathbb{E}_{\gamma_{Z|(\hat{X}, \hat{Z}, X)}} [\cdot | \hat{X}, \hat{Z}, X] | \hat{X}, X] | \hat{X}]] \\ &= \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} [\mathbb{E}_{\gamma_{X|\hat{X}}} [\mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\mathbb{E}_{\gamma_{Z|(\hat{X}, \hat{Z}, X)}} [\cdot | \hat{X}, \hat{Z}, X] | \hat{X}, X] | \hat{X}]], \end{aligned}$$

where we use $\gamma_{\hat{Z}|\hat{X}, X} = \hat{\mathbb{P}}_{\hat{Z}|\hat{X}}$ for almost every $(\hat{X}, X)$ because γ is causal. Therefore, we have

$$\begin{aligned} &\mathbb{E}_{\gamma} [\Psi(f(X), Z) - \lambda \|X - \hat{X}\|^p - \lambda \|Z - \hat{Z}\|^p] \\ &= \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} [\mathbb{E}_{\gamma_{X|\hat{X}}} [\mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\mathbb{E}_{\gamma_{Z|(\hat{X}, \hat{Z}, X)}} [\Psi(f(X), Z) - \lambda \|X - \hat{X}\|^p \\ &\quad - \lambda \|Z - \hat{Z}\|^p | \hat{X}, \hat{Z}, X] | \hat{X}, X] | \hat{X}]] \\ &\leq \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} \left[\sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda \|z - \hat{Z}\|^p \} \right] \right\} \right. \\ &\quad \left. - \lambda \|x - \hat{X}\|^p \right\} \right]. \end{aligned}$$

This completes the proof of the weak duality. $\square$

The strong duality result states as follows.

Theorem 1 (Strong Duality). *Let $f : \mathcal{X} \rightarrow \mathcal{D}$ be a measurable function. Then $v_P^f = v_D^f$.*

Proof Sketch. The proof idea is to construct a nearly worst-case distribution of the primal problem based on the first-order optimality condition of the weak dual problem (D^f) . Conceptually, the duality of Causal transport DRO shares common ideas with the duality proof of Wasserstein DRO (Gao and Kleywegt 2023): Both establish duality by constructing a nearly worst-case scenario via the first-order optimality conditions of the weak dual problem. However, the construction in our setting differs in several key aspects. Unlike standard Wasserstein DRO—where the decision variable typically interacts symmetrically with the joint distribution—our formulation requires handling the asymmetric roles of the variables X and Z under causally ordered constraints. This asymmetry introduces new technical challenges in characterizing admissible perturbations and conditional transport maps. As a result, the worst-case distribution in our framework involves a two-layer construction: first over marginal perturbations and then over conditional distributions. The dual characterization must therefore simultaneously account for both layers while ensuring measurability and feasibility under causal constraints. This layered structure, combined with the topological and measurability requirements inherent in causal transport, makes our duality proof technically nontrivial and distinct from existing formulations.

The worst-case distribution maximizes the expected loss within a given transport budget. With a fixed dual variable λ , the worst-case distribution for the soft constraint problem

$$\max_{\mathbb{P} \in \mathcal{P}(\mathcal{X} \times \mathcal{Z})} \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] - \lambda C_p(\hat{\mathbb{P}}, \mathbb{P})^p$$

is obtained by moving $\hat{z}_{ki}$ toward the maximizer of the innermost maximization problem of (D^f)

$$\Upsilon(\lambda; x, \hat{z}_{ki}) := \sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda \|z - \hat{z}_{ki}\|^p \},$$

and moving $\hat{x}_k$ toward the maximizer of the maximization problem

$$\sup_{x \in \mathcal{X}} \{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\Upsilon(\lambda; x, \hat{Z}) | \hat{X} = \hat{x}_k] - \lambda \|x - \hat{x}_k\|^p \}.$$

One can see that such a transport plan is causal: The maximizer x is the perturbation of $\hat{x}_k$ in the transport plan, and x is solely determined by $\hat{x}_k$, independent of which $\hat{z}_{ki}$ it is associated with. If both maximizers over x and over z exist and are unique at the critical λ^* dual to the given transport distance ρ^p , then the transport plan would induce a worst-case distribution. If the maximizer does not exist or is not unique, two alternative transport plans are considered: One produces a

feasible but suboptimal distribution, and the other, although infeasible, achieves a higher objective value. Interpolating between these distributions allows for a near-optimal solution to the primal problem.

As can be seen from the definition of Υ , the worst-case distribution for the soft constraint problem is obtained by moving mass with loss-to-distance “efficiency” higher than λ . Efficiency here refers to the ratio of gain (or loss reduction) to the p th power of distance. Specifically, the efficiency of moving $\hat{x}$ to x is $(\mathbb{E}[\Upsilon(\lambda; x, \hat{Z})] - \mathbb{E}[\Upsilon(\lambda; \hat{x}, \hat{Z})]) / \|x - \hat{x}\|^p$, in which Υ already incorporated the efficiency of moving $\hat{z}$ to z , calculated by $(\Psi(f(x), z) - \Psi(f(x), \hat{z})) / \|z - \hat{z}\|^p$.

There are several possibilities where the near-optimal distribution is located, depending on the critical threshold λ^* that minimizes (D^f). Indeed, the dual objective function

$$h(\lambda) := \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|x}} \left[\sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda \|z - \hat{Z}\|^p \} \right] \right. \right. \\ \left. \left. - \lambda \|x - \hat{X}\|^p \right\} \right] \quad (5)$$

is an extended-real-valued convex function of λ ; $h(\lambda) - \lambda \rho^p$ is monotonically decreasing in λ and coincides with the above soft constraint problem. Let $\kappa \in [0, +\infty]$ be the smallest value such that the dual objective is finite in $(\kappa, +\infty)$. The infimum over λ in (D^f) can have several possibilities:

- Case 1: $\kappa = +\infty$, so the dual objective is $+\infty$ for any $\lambda > 0$. This means that by transporting an arbitrarily small distance, one can generate an arbitrarily large loss.
- Case 2: $\kappa < +\infty$ and minimization over λ in (D^f) is achieved in the interior of $(\kappa, +\infty)$. The dual objective can be arbitrarily large if λ is smaller than κ , but it would require transporting mass that exhausts the transport distance budget. We interpolate two transport plans: moving all the masses with “efficiency” above $\lambda_1 < \lambda^*$ (superoptimal but infeasible) versus moving all the masses with efficiency above $\lambda_2 > \lambda^*$ (feasible but suboptimal).
- Case 3: $\kappa < +\infty$ and v_{WD}^f is minimized at κ . Moving all the mass with efficiency strictly above κ does not exhaust the transport distance budget. This is further divided into the following:

□ Case 3.1: $\kappa = 0$. Any positive λ corresponds to a finite soft loss. We simply move all the mass with positive efficiency.

□ Case 3.2: $\kappa > 0$. We again interpolate two transport plans: moving all the masses with efficiency above $\lambda_2 > \lambda^*$ (feasible but suboptimal) versus moving *some* of the masses with efficiency above $\kappa_1 < \lambda^*$ (superoptimal but infeasible). We can only move the latter up to some distance, in contrast to Case 2, because moving them all would travel an infinite distance.

We refer to the next section for a more detailed construction of a worst-case distribution and Online Appendix EC.3 for a complete proof. □

Remark 1 (Comparison with Wasserstein DRO). Recall the Wasserstein DRO problem

$$\sup_{\mathbb{P}} \{ \mathbb{E}_{(X,Z) \sim \mathbb{P}} [\Psi(f(X), Z)] : \mathcal{W}_p(\hat{\mathbb{P}}, \mathbb{P}) \leq \rho \},$$

which has the following equivalent dual form (Gao and Kleywegt 2023, Zhang et al. 2025):

$$v_{\text{WD}}^f := \inf_{\lambda \geq 0} \left\{ \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}} \left[\sup_{\substack{x \in \mathcal{X} \\ z \in \mathcal{Z}}} \{ \Psi(f(x), z) - \lambda \|z - \hat{Z}\|^p \} \right. \right. \\ \left. \left. - \lambda \|x - \hat{X}\|^p \right\} \right] \\ = \inf_{\lambda \geq 0} \left\{ \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) \} \right. \right. \right. \right. \\ \left. \left. \left. - \lambda \|z - \hat{Z}\|^p \right\} - \lambda \|x - \hat{X}\|^p \right] \right] \right\}.$$

Comparing it with the dual problem (D^f) of causal transport distance DRO, the difference is the swap of supremum over x and the conditional expectation of $\hat{Z}$ given $\hat{X}$. Hence, if the switching does not change the objective value, which holds, for instance, when the conditional distribution $\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}$ is a Dirac measure for every $\hat{X}$, then the Wasserstein DRO dual problem and causal transport distance DRO dual problems are equal. From a primal point of view, if $\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}$ is Dirac for every $\hat{X}$, then every transport plan from $\hat{\mathbb{P}}$ to $\mathbb{P}$ is causal. In this case, the causal transport distance DRO and Wasserstein DRO coincide. Intuitively, if every conditional distribution $\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}$ is Dirac, then the nominal distribution does not have any meaningful conditional information structure to exploit, and thus the causal transport distance DRO reduces to Wasserstein DRO.

Without considering the causality constraint, the optimal strategy is the *greedy* one. When a unit of mass is moved from $(\hat{x}, \hat{z})$ to (x, z) , it generates a “revenue” of $\Psi(f(x), z) - \Psi(f(\hat{x}), \hat{z})$ while incurring a transport distance $\|x - \hat{x}\|^p + \|z - \hat{z}\|^p$. Here, the “revenue” refers to the gain (increase in the objective) relative to the cost of moving the mass. The efficiency of this transportation is thus $(\Psi(f(x), z) - \Psi(f(\hat{x}), \hat{z})) / (\|x - \hat{x}\|^p + \|z - \hat{z}\|^p)$. It will move $(\hat{x}, \hat{z})$ to a destination with the (near-) highest efficiency, and $(\hat{x}, \hat{z})$ is moved only after all other sources $(\hat{x}', \hat{z}')$ with higher efficiency have been depleted. This greedy strategy is reflected in v_{WD}^f . The dual objective computes the net profit of transporting all the mass with efficiency higher than threshold λ with transport cost multiplied by a factor of λ (toll rate), and v_{WD}^f computes the revenue by reimbursing the transport cost $\lambda \rho^p$ and then searches for the critical threshold λ^* .

3.2. Worst-Case Distribution

In this section, we investigate the structure of the worst-case distribution and its existence conditions. Compared with the results in Section 3.1, in the following result, we require $\mathcal{X}$ and $\mathcal{Z}$ to be finite-dimensional and thus locally compact and require some continuity assumptions on Ψ so that the maximizers are attainable.

Theorem 2 (Worst-Case Distribution). *Suppose $\mathcal{X}, \mathcal{Z}$ are finite dimensional, and $\Psi(f(\cdot), \cdot)$ is upper semicontinuous. If the optimal value of (D^f) is attained at some $\lambda^* > \kappa$ for κ specified in Lemma EC.2 in the Online Appendix, then a worst-case distribution exists and has the following form:*

$$\mathbb{P}^* = \sum_{k \neq k_0} \sum_{i=1}^{n_k} \hat{p}_{ki} \delta_{(x_k^*, z_{ki}^*)} + \sum_{i=1}^{n_{k_0}} \hat{p}_{k_0 i} (q \delta_{(\bar{x}_{k_0}, \bar{z}_{k_0 i})} + (1-q) \delta_{(\underline{x}_{k_0}, \underline{z}_{k_0 i})}),$$

where $1 \leq k_0 \leq K$, $0 \leq q \leq 1$, $(x_k^*, z_{ki}^*) = (\bar{x}_k, \bar{z}_{ki})$, and for every k and i ,

$$\begin{aligned} \bar{x}_k, \underline{x}_k &\in \arg \max_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\mathcal{Z}|X}} \left[\sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda^* \|z - \hat{z}\|^p \} \mid \hat{X} = \hat{x}_k \right] - \lambda^* \|x - \hat{x}_k\|^p \right\}, \\ \bar{z}_{ki} &\in \arg \max_{z \in \mathcal{Z}} \{ \Psi(f(\bar{x}_k), z) - \lambda^* \|z - \hat{z}_{ki}\|^p \}, \\ \underline{z}_{ki} &\in \arg \max_{z \in \mathcal{Z}} \{ \Psi(f(\underline{x}_k), z) - \lambda^* \|z - \hat{z}_{ki}\|^p \}. \end{aligned}$$

From Theorem 2, we see that there exists a worst-case distribution $\mathbb{P}^*$ supported on at most $N + n_{k_0}$ points, and its marginal $\mathbb{P}_X^*$ is supported on at most $K + 1$ points. We demonstrate the structure of the worst-case distribution in Figure 3 (left). In this plot, the support of $\hat{\mathbb{P}}$ is represented by $\bullet$, and we have $K = 3$, $n_k = 3$, $k = 1, 2, 3$, and $k_0 = 2$. These points are transported to $\star$ s, which form the worst-case distribution $\mathbb{P}^*$. For $k = 1, 3$, we observe that $\hat{x}_k$ is transported to x_k^* , and the conditional distribution $\mathbb{P}_{\mathcal{Z}|X=x_k^*}^*$ has the same structure as the conditional distribution $\hat{\mathbb{P}}_{\mathcal{Z}|X=\hat{x}_k}$, both supported on three points with identical probability mass function $(\hat{p}_{ki})_{i=1,2,3}$. Furthermore, $\hat{x}_2$ is split into two values $\bar{x}_2$

and $\underline{x}_2$, and the conditional distributions $\mathbb{P}_{\mathcal{Z}|X=\bar{x}_2}^*$, $\mathbb{P}_{\mathcal{Z}|X=\underline{x}_2}^*$ have the same structure as the conditional distribution $\hat{\mathbb{P}}_{\mathcal{Z}|X=\hat{x}_2}$, both supported on three points with identical probability mass function $(\hat{p}_{2i})_{i=1,2,3}$.

As a comparison, on the right side of Figure 3, we plot the worst-case distribution resulting from Wasserstein DRO. According to Gao and Kleywegt (2023), the worst-case distribution can be supported on $N + 1$ points, and points with the same x -value could have different x -values after transportation or splitting. The conditional distributions of the worst-case distribution change completely, each of which is a Dirac measure. This example illustrates that the worst-case distribution of the causal transport distance DRO preserves the conditional information structure of the nominal distribution, whereas the Wasserstein DRO fails to do so.

We illustrate the worst-case distributions under Wasserstein DRO and causal transport DRO for the mean estimation problem as follows.

Example 4 (Revisited). Consider the conditional mean estimation problem in Example 4. We compare the worst-case distributions with 2-Wasserstein DRO and 2-causal transport DRO when the decision rule $f = f_{\text{true}}$ is the true conditional mean, and the uncertainty set radius is $\rho = 0.2$. As shown in Figure 4, in the worst-case Wasserstein DRO, the conditional information structure is not preserved.

4. Finding the Optimal Decision Rule

In this section, we study the outer optimization over decision rules in (P). As a direct consequence of Theorem 1, Problem (P) is equivalent to the following:

$$v_D := \inf_{\substack{f \in \mathcal{F} \\ \lambda \geq 0}} \left\{ \lambda \rho^p + \mathbb{E}_{\mathbb{P}_X} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\mathcal{Z}|X}} \left[\sup_{z \in \mathcal{Z}} \{ \Psi(f(x), z) - \lambda \|z - \hat{z}\|^p \} \mid \hat{X} = x \right] - \lambda \|x - \hat{X}\|^p \right\} \right] \right\}. \quad (D)$$

In particular, if we define $\|z - \hat{z}\|_{\mathcal{Z}} := \infty \mathbf{1}\{z \neq \hat{z}\}$, which is often used when the side information is relatively

Figure 3. (Color online) Structure of the Worst-Case Distributions

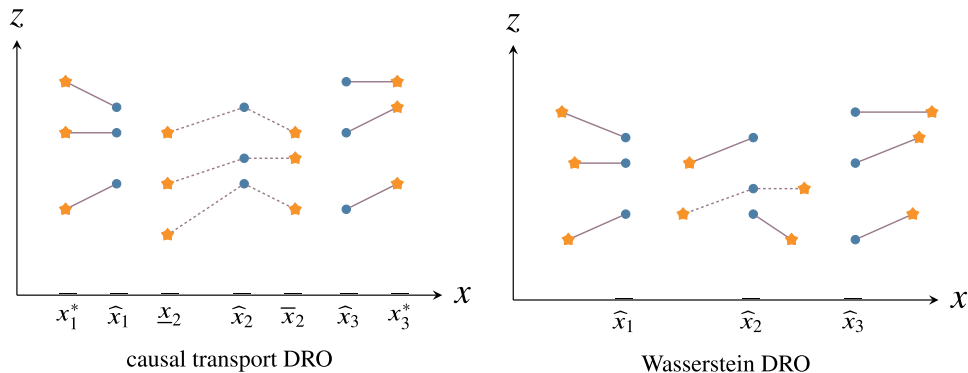
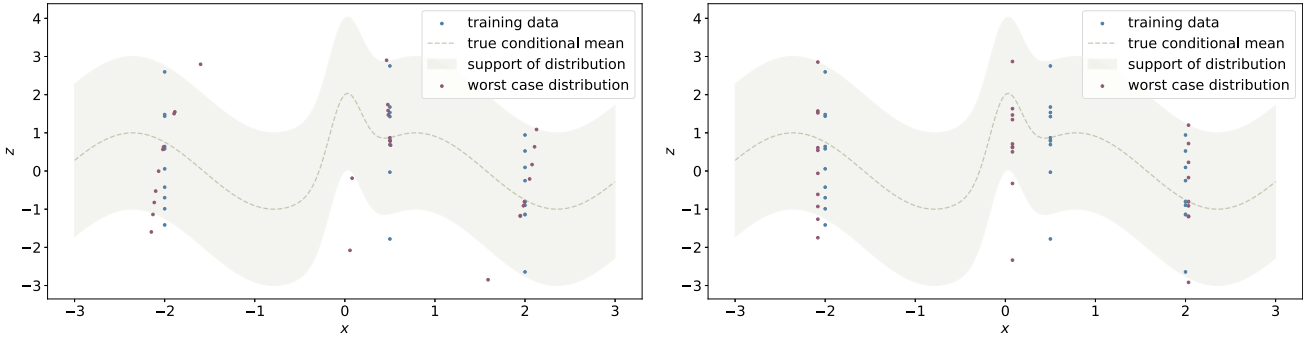


Figure 4. (Color online) Structure of the 2-Wasserstein (Left) vs. Causal (Right) Worst-Case Distributions for Mean Estimation



accurate, then (D) is simplified to

$$v_D := \inf_{\substack{f \in \mathcal{F} \\ \lambda \geq 0}} \left\{ \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\Psi(f(x), \hat{Z}) | \hat{X}] - \lambda \|x - \hat{X}\|^p \right\} \right] \right\}. \quad (6)$$

The tractability of the optimization over $f \in \mathcal{F}$ depends on the class of decision rules $\mathcal{F}$. If $\mathcal{F}$ admits a finite-dimensional parameterization, such as an affine class, then Problem (D) is a finite-dimensional optimization, and we identify cases where the overall problem can be solved by off-the-shelf convex programming solvers (Section 4.1). Otherwise, if $\mathcal{F}$ is a nonparametric class, and particularly the class of all decision rules, then the optimization over $\mathcal{F}$ is an infinite-dimensional functional optimization; yet still, we identify cases where the overall problem can be solved efficiently (Section 4.3).

4.1. Optimizing over Affine Decision Rules

In this section, we provide tractable formulations when $\mathcal{F}$ is an affine class. Suppose affine functions in $\mathcal{F}$ are parametrized by Θ :

$$\mathcal{F}_{\Theta} = \{x \mapsto B^{\top}x + \delta : (B, \delta) \in \Theta\}, \quad (7)$$

where Θ is a finite-dimensional convex set.

Our first result shows that (6) is tractable when Ψ is affine in the decision variable w . The proof can be found in Online Appendix EC.4.

Corollary 1. Suppose $\mathcal{F} = \mathcal{F}_{\Theta}$ as defined in (7), and $\Psi(\cdot, z)$ is affine for every z ; that is, there exist functions $\beta(\cdot), b(\cdot)$ such that

$$\Psi(w, z) = \beta(z)^{\top}w + b(z).$$

Set

$$\hat{p}_k := \sum_{i=1}^{n_k} \hat{p}_{ki}, \quad \beta_k := \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\beta(\hat{Z}) | \hat{X} = \hat{x}_k],$$

$$b_k := \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [b(\hat{Z}) | \hat{X} = \hat{x}_k].$$

Then, the dual problem (6) is equivalent to the following convex programs. When $p = 1$, (6) is equivalent to

$$\inf_{(B, \delta) \in \Theta} \rho^p \cdot \max_{k \in [K]} \|B\beta_k\|_* + \sum_{k=1}^K \hat{p}_k (\beta_k^{\top} (B^{\top} \hat{x}_k + \delta) + b_k).$$

When $p \in (1, +\infty)$, (6) is equivalent to

$$\inf_{\lambda \geq 0, (B, \delta) \in \Theta} \lambda \rho^p + \sum_{k=1}^K \hat{p}_k \left(\beta_k^{\top} (B^{\top} \hat{x}_k + \delta) + b_k + \lambda(p-1) \left(\frac{\|B\beta_k\|_*}{\lambda p} \right)^{\frac{p}{p-1}} \right).$$

Here $\|\cdot\|_*$ is the dual norm of $\|\cdot\|_{\mathcal{X}}$.

As a special case, we assume further that $\Psi(w, z)$ is bilinear. When $p = 2$, the above convex program can be written as a positive semidefinite program.

Corollary 2. Suppose $\mathcal{F} = \mathcal{F}_{\Theta}$ as defined in (7) and $\Psi(w, z)$ is bilinear:

$$\Psi(w, z) = w^{\top}Az + \beta^{\top}w + \alpha^{\top}z + b.$$

Set

$$\hat{p}_k = \sum_{i=1}^{n_k} \hat{p}_{ki}, \quad \bar{z}_k = \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|\hat{X}}} [\hat{Z} | \hat{X} = \hat{x}_k]. \quad (8)$$

Then (D) with $p = 2$ is equivalent to the following positive semidefinite program:

$$\inf_{\substack{(B, \delta) \in \Theta \\ \lambda \geq 0, \{y_k\}_k \subset \mathbb{R}}} \lambda \rho^2 + \sum_{k=1}^K \hat{p}_k y_k$$

$$\text{s.t.} \quad \begin{pmatrix} \lambda I & -\frac{1}{2}BA & -\frac{1}{2}B\beta - \lambda \hat{x}_k \\ -\frac{1}{2}(BA)^{\top} & \lambda I & -\frac{1}{2}(A^{\top}\delta + \alpha) - \lambda \bar{z}_k \\ -\frac{1}{2}(B\beta)^{\top} - \lambda \hat{x}_k^{\top} & -\frac{1}{2}(A^{\top}\delta + \alpha)^{\top} & y_k - \beta^{\top}\delta - b \\ & -\lambda \bar{z}_k^{\top} & + \lambda \|\bar{z}_k\|^2 + \lambda \|\hat{x}_k\|^2 \end{pmatrix} \succeq O, \quad k \in [K].$$

Here O stands for the zero matrix and I represents the identity matrix.

Example 7 (Revisited). We revisit the contextual linear optimization problem in Example 7, where the decision is restricted to a polygon $\mathcal{D} = \{w \in \mathbb{R}^m : Cw \leq c\}$, and the context X is bounded in an ellipsoid $\mathcal{X} = \{x \in \mathbb{R}^d : (x - x_0)^\top \Sigma (x - x_0) \leq R\}$. Here $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric and positive definite, $x_0 \in \mathbb{R}^d$, $R > 0$, $C \in \mathbb{R}^{L \times m}$, and $c \in \mathbb{R}^L$. $Cw \leq c$ means $C_\ell^\top w \leq c_\ell$, where $C_\ell^\top$ is the ℓ th row of C and c_ℓ is the ℓ th entry of c , for each $\ell \in [L]$. Θ defined by (4) is convex. Using Corollary 2, (D) with $p = 2$ can be reformulated as the following positive semidefinite program:

$$\begin{aligned} \inf_{\substack{B \in \mathbb{R}^{d \times m}, \delta \in \mathbb{R}^m \\ \lambda \geq 0, \{y_k\} \in \mathbb{R}^K \\ \{\mu_k\}_k \geq 0, \{v_\ell\}_\ell \geq 0}} \quad & \lambda \rho^2 + \sum_{k=1}^K \hat{p}_k y_k \\ \text{s.t.} \quad & \begin{pmatrix} \lambda I + \mu_k \Sigma & -\frac{1}{2}B & -\lambda \hat{x}_k - \mu_k \Sigma x_0 \\ -\frac{1}{2}B^\top & \lambda I & -\frac{1}{2}\delta - \lambda \bar{z}_k \\ -\lambda \hat{x}_k^\top - \mu_k (\Sigma x_0)^\top & -\frac{1}{2}\delta^\top - \lambda \bar{z}_k^\top & y_k + \lambda \|\bar{z}_k\|^2 + \lambda \|\hat{x}_k\|^2 \\ & & + \mu_k x_0^\top \Sigma x_0 - \mu_k R \end{pmatrix} \succeq 0 \\ & \forall k \in [K], \\ & \begin{pmatrix} v_\ell \Sigma & -\frac{1}{2}BC_\ell - v_\ell \Sigma x_0 \\ -\frac{1}{2}C_\ell^\top B^\top - v_\ell (\Sigma x_0)^\top & -C_\ell^\top \delta + c_\ell + v_\ell x_0^\top \Sigma x_0 - v_\ell R \end{pmatrix} \succeq 0 \\ & \forall \ell \in [L]. \end{aligned}$$

Recall $\hat{p}_k$ and $\bar{z}_k$ are defined in (8). Detailed computations are provided in Online Appendix EC.5.

Our second result shows (D) is tractable when the loss function Ψ is quadratic and $p = 2$. We study Example 4 in detail, and we provide the proof in Online Appendix EC.5.

Corollary 3. We revisit the conditional mean estimation problem in Example 4. Assume $\mathcal{X} = \mathbb{R}^d$, $\mathcal{Z} = \mathbb{R}^m$, and $\Theta = \mathbb{R}^{d \times m} \times \mathbb{R}^m$. Each space is equipped with the Euclidean norm ℓ^2 . With quadratic loss $\Psi(w, z) = \|w - z\|^2$ and affine policy $f(x) = B^\top x + \delta$, the dual problem (D) can be formulated as the following positive semidefinite program:

$$\begin{aligned} \inf_{\substack{(B, \delta) \in \Theta \\ \lambda \geq 1, t \geq 0 \\ \{y_k\} \in \mathbb{R}^K \\ W \in \mathbb{R}^{d \times d}, u \in \mathbb{R}^d}} \quad & \lambda \rho^2 + \lambda \sum_k \hat{p}_k y_k \\ \text{s.t.} \quad & \begin{pmatrix} (\lambda - 1)I - W & -(\lambda - 1)\hat{x}_k + B\mu_k - u \\ -(\lambda - 1)\hat{x}_k^\top & (\lambda - 1)(y_k + |\hat{x}_k|^2) \\ + \mu_k^\top B^\top - u^\top & -t + 2\delta^\top \mu_k - |\mu_k|^2 - \sigma_k \end{pmatrix} \succeq 0 \\ & \forall k \in [K], \\ & \begin{pmatrix} W & B & u \\ B^\top & I & \delta \\ u^\top & \delta^\top & t \end{pmatrix} \succeq 0. \end{aligned}$$

Here $\mu_k = \mathbb{E}[\hat{Z} | \hat{X} = \hat{x}_k]$ and $\sigma_k = \text{Var}[\hat{Z} | \hat{X} = \hat{x}_k]$ represent the conditional mean and the conditional variance, respectively.

4.2. Optimizing over Reproducing Kernel Hilbert Space Decision Rules

Restricting decision rules to affine functions of the contexts, as in Section 4.1, can be suboptimal. To address this limit, in this section, we consider the decision rule space $\mathcal{F}$ to be a vector-valued reproducing kernel Hilbert space (RKHS; Carmeli et al. 2010). This class of policies can be interpreted as affine functions applied to nonlinear transformations of the contexts, where the nonlinear transformations are specified implicitly through a kernel function. The kernel function measures similarity between contexts and is often more interpretable than explicitly specifying the nonlinear transformation itself. Moreover, when the kernel function is universal, the corresponding RKHS policies are asymptotically optimal.

Specifically, let $\mathcal{F} = \mathbb{H} \subset \{\mathbb{h} : \mathcal{X} \rightarrow \mathbb{R}^m\}$ be a vector-valued RKHS equipped with an inner product $\langle \cdot, \cdot \rangle_{\mathbb{H}}$ and a kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}^{m \times m}$, such that the following reproducing property holds: The j th component of $\mathbb{h}$ can be expressed by

$$\mathbb{h}_j(x) = \langle \mathbb{h}, K(\cdot, x)e_j \rangle_{\mathbb{H}}, \quad \forall j \in [m], x \in \mathcal{X},$$

where e_j is the j th unit vector of $\mathbb{R}^m$. Define a map $\varphi : \mathcal{X} \rightarrow \mathbb{H}^m$ via $\varphi_j(x) := K(\cdot, x)e_j$, $j \in [m]$, which lifts the state space $\mathcal{X}$ to the feature space $\mathbb{H}^m$. Under this definition, the above expression can be written as

$$\mathbb{h}(x) = \langle \mathbb{h}, \varphi(x) \rangle_{\mathbb{H}} := \{\langle \mathbb{h}, \varphi_j(x) \rangle_{\mathbb{H}}\}_{j \in [m]},$$

and the true loss of a policy h can be rewritten as

$$\begin{aligned} \mathbb{E}_{(X, Z) \sim \mathbb{P}_{\text{true}}} [\Psi(\mathbb{h}(X), Z)] &= \mathbb{E}_{(X, Z) \sim \mathbb{P}_{\text{true}}} [\Psi(\langle \mathbb{h}, \varphi(X) \rangle_{\mathbb{H}}, Z)] \\ &= \mathbb{E}_{(\Phi, Z) \sim \mathbb{Q}_{\text{true}}} [\Psi(\langle \mathbb{h}, \Phi \rangle_{\mathbb{H}}, Z)], \end{aligned}$$

where $\mathbb{Q}_{\text{true}} = (\varphi \oplus \text{id}_{\mathcal{Z}})_{\#} \mathbb{P}_{\text{true}} \in \mathcal{P}(\mathbb{H}^m \times \mathcal{Z})$ is the true joint distribution between the lifted feature Φ and the random parameter Z . Recall that $\#$ denotes the push-forward of a measure, and the direct sum $\varphi \oplus \text{id}_{\mathcal{Z}} : \mathcal{X} \times \mathcal{Z} \rightarrow \mathbb{H}^m \times \mathcal{Z}$ is defined as $(\varphi \oplus \text{id}_{\mathcal{Z}})(x, z) := (\varphi(x), z)$.

We now define the causal transport robust formulation, following a way similar to the RKHS-based 1-Wasserstein uncertainty sets proposed in Shafieezadeh-Abadeh et al. (2019). Define the nominal distribution as $\hat{\mathbb{Q}} := (\varphi \oplus \text{id}_{\mathcal{Z}})_{\#} \hat{\mathbb{P}}$, representing the joint nominal distribution of the feature Φ and the random parameter Z . For $\rho > 0$, define the 1-causal transport uncertainty set on the space of feature distributions:

$$\mathfrak{M} = \{\mathbb{Q} \in \mathcal{P}(\mathbb{H}^m \times \mathcal{Z}) : C_1(\hat{\mathbb{Q}}, \mathbb{Q}) \leq \rho\}.$$

We consider the following problem:

$$v_{\mathcal{P}} = \inf_{\mathbb{h} \in \mathbb{H}} \max_{\mathbb{Q} \in \mathfrak{M}} \mathbb{E}_{(\Phi, Z) \sim \mathbb{Q}} [\Psi(\langle \mathbb{h}, \Phi \rangle_{\mathbb{H}}, Z)]. \quad (9)$$

Note that both the inf and max problems are infinite-dimensional, because they are optimizing over functions and probability distributions, respectively.

We derive the following finite-dimensional convex reformulation of (9). For simplicity, we equip $\mathcal{Z}$ with infinity metric $d(z, z') = \infty \mathbf{1}\{z \neq z'\}$.

Corollary 4. Suppose $\Psi(\cdot, z)$ is convex and Lipschitz in the decision variable. Define a constant

$$L := \sup_{\hat{x} \in \text{supp } \hat{\mathbb{P}}_{\hat{X}}} \|\mathbb{E}_{\hat{Z} \sim \hat{\mathbb{P}}_{\hat{Z}} | \hat{X}} [\Psi(\cdot, \hat{Z}) | \hat{X} = \hat{x}]\|_{\text{Lip}}.$$

Then $v_{\mathbb{P}}$ equals

$$v_{\mathbb{D}} = \min_{\{\beta_k\}_k \subset \mathbb{R}^m} \left\{ \mathbb{E}_{(\hat{X}, \hat{Z}) \sim \hat{\mathbb{P}}} \left[\Psi \left(\sum_{k \in [K]} K(\hat{X}, \hat{x}_k) \beta_k, \hat{Z} \right) \right] + \rho L \left(\sum_{j, k \in [K]} \beta_j^T K(\hat{x}_j, \hat{x}_k) \beta_k \right)^{\frac{1}{2}} \right\}. \quad (10)$$

Moreover, the optimal h that solves (9) is given by

$$\mathbb{h}^* = \sum_{k \in [K]} K(\cdot, \hat{x}_k) \beta_k^*,$$

where β^* is the minimizer of (10).

This result demonstrates the tractability of the robust formulation (9). A key computational advantage is that it reduces the infinite-dimensional minimax problem to a minimization over only mK real variables, with the robust loss naturally serving as a norm regularization. The proof is provided in Online Appendix EC.4. At a high level, the argument builds on the proof of Shafieezadeh-Abadeh et al. (2019, theorem 28), which applies the representer theorem for RKHS, and extends the analysis to vector-valued functions and the causal transport distance.

4.3. Optimizing over All (Nonparametric) Decision Rules

In this section, we consider $\mathcal{F}$ to be unrestricted and contain all measurable functions $\{f: \mathcal{X} \rightarrow \mathcal{D}\}$. In general, this infinite-dimensional problem is hard to solve. Nonetheless, we provide a tractable way to find the optimal decision rule for this problem in certain settings.

Recall that our dual reformulation in Theorem 1 states that

$$v_{\mathbb{D}} = \min_{f: \mathcal{X} \rightarrow \mathcal{D}} \min_{\lambda \geq 0} \left\{ \lambda \rho + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \{\varphi(f(x); \lambda, \hat{X}) - \lambda \|x - \hat{X}\|\} \right] \right\}, \quad (11)$$

where $\varphi(w; \lambda, \hat{x}) := \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}} | \hat{X}} [\sup_{z \in \mathcal{Z}} \{\Psi(w, z) - \lambda \|z - \hat{Z}\|\} | \hat{X} = \hat{x}]$. By replacing $\mathcal{X}$ with $\text{supp } \hat{\mathbb{P}}$, we define the in-sample

dual problem as

$$v_{\mathbb{D}} := \min_{\substack{f: \mathcal{X} \rightarrow \mathcal{D} \\ \lambda \geq 0}} \left\{ \lambda \rho + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\max_{1 \leq k \leq K} \{\varphi(f(\hat{x}_k); \lambda, \hat{X}) - \lambda \|\hat{x}_k - \hat{X}\|\} \right] \right\} \quad (12)$$

$$= \min_{\substack{f \in \mathcal{F} \\ \lambda \geq 0}} \left\{ \lambda \rho + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\max_{1 \leq k \leq K} \{\varphi(\hat{f}(\hat{x}_k); \lambda, \hat{X}) - \lambda \|\hat{x}_k - \hat{X}\|\} \right] \right\}, \quad (13)$$

where the second equality holds because the objective value in (12) depends only on the value of f on $\text{supp } \hat{\mathbb{P}}$. Note that (13) is a finite-dimensional convex optimization problem with $K + 1$ decision variables in the outer minimization.

Theorem 3. Suppose $p = 1$, $\mathcal{D} \subset \mathbb{R}$ is convex, and $\Psi(w, z)$ is convex in w . Let $(\lambda^*, \hat{f}^*)$ be a minimizer to the in-sample dual problem (13). Denote $\varphi_k(w) := \varphi(w; \lambda^*, \hat{x}_k)$, $w_k := \hat{f}^*(\hat{x}_k)$, and $\phi_k := \max_j \{\varphi_k(w_j) - \lambda^* \|\hat{x}_k - \hat{x}_j\|\}$. For $x \in \mathcal{X}$, define

$$I_k(x) := \{w \in \mathcal{D} : \varphi_k(w) \leq \lambda^* \|x - \hat{x}_k\| + \phi_k\}.$$

Then the intersection of $I_k(x)$ s is nonempty, and every decision rule $f^* \in \mathcal{F}$ satisfying $f^*(x) \in \cap_k I_k(x)$ for all $x \in \mathcal{X}$ is a minimizer to (11). Moreover, let (λ^*, f^*) be a minimizer to the dual problem (D), then $(\lambda^*, \hat{f}^*)$ is a minimizer to (13), and $f^*(x) \in \cap_k I_k(x)$ defined above.

Theorem 3 shows that Problems (11) and (13) share the same optimal dual variable λ^* , and to solve the infinite-dimensional optimization over decision rules (11), it suffices first to solve a finite-dimensional robust in-sample optimization (13) and then extend the robust optimal in-sample decision rule to $\mathcal{X} \setminus \text{supp } \hat{\mathbb{P}}$, such that it is optimal to the original problem. Note that once the in-sample problem (13) is solved, the values w_k, ϕ_k are immediately available, and the set I_k is defined precisely. There may be more than one way to extend the in-sample robust optimal decision rule $\hat{f}$ to the entire space, as long as it belongs to the range of $\cap_k I_k(x)$.

Proof Sketch. The idea behind proving Theorem 3 is as follows. To show the optimality of the decision rules that lie within the intersection $\cap_k I_k$, the key step is to show $v_{\mathbb{D}} = v_{\mathbb{D}}$. Observe that $v_{\mathbb{D}} \geq v_{\mathbb{D}}$, because the inner supremum in (11) is taken with respect to a larger set compared with the maximization in (12). To see the other direction, the main step is to show that $I_k(x)$ has a nonempty intersection. Once this is shown, it is easy to verify by simple algebra that $f^*(x) \in \cap_k I_k(x)$ attains the value $v_{\mathbb{D}}$; thereby, $v_{\mathbb{D}}$ is dominated by the objective value of f^* that equals $v_{\mathbb{D}}$. Thus, we have $v_{\mathbb{D}} = v_{\mathbb{D}}$. To show $I_k(x)$ has a nonempty intersection, because they are one-dimensional intervals, it suffices to show they pairwise intersect. This can be established using the convexity of φ . The necessity of the above interval

condition, that is, for any optimal policy f^* , $f^*(x) \in \cap_k I_k(x)$, could be justified by contradiction. The detailed proof can be found in Online Appendix EC.4. $\square$

Remark 2. The one-dimensional assumption of the decision space $\mathcal{D}$ is crucial. The following example shows that it is impossible to use the form in Theorem 3 to construct an optimal policy as soon as the decision dimension becomes two. Specifically, let us consider a two-dimensional decision space $\mathcal{D}$ and a linear loss function $\Psi(w, z) = w^\top z$, where $\mathcal{D} = B_1$ is the unit disk in $\mathbb{R}^2$, and $\mathcal{X} = \mathcal{Z} = \mathbb{R}^2$. Suppose $\hat{\mathbb{P}}$ satisfies

$$\hat{\mathbb{P}}_{\hat{X}} = \frac{1}{3} \sum_{k=1}^3 \delta_{\hat{x}_k}, \quad \hat{x}_k = \left(\cos \frac{2k\pi}{3}, \sin \frac{2k\pi}{3} \right),$$

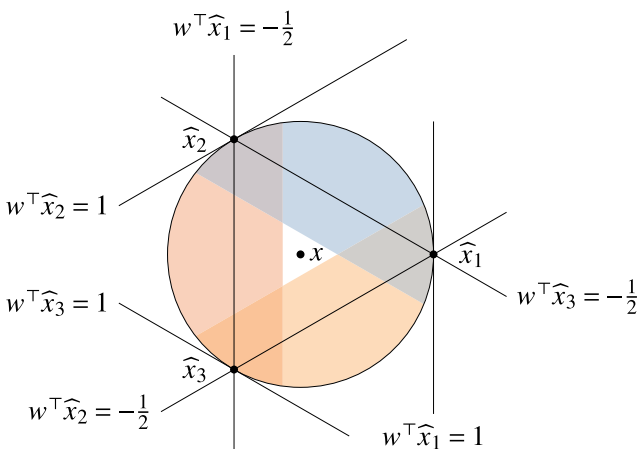
$$\mathbb{E}[\hat{Z} | \hat{X}] = \frac{2\sqrt{3}}{3} \hat{X},$$

where $\hat{X} = \{\hat{x}_1, \hat{x}_2, \hat{x}_3\}$ consists of three points on the unit circle that form an equilateral triangle (Figure 5). We can show that the intersection of $I_k(x)$ defined in Theorem 3 is empty, although the in-sample dual problem is explicitly solvable. Detailed calculations are available in Online Appendix EC.4.

Remark 3 (Comparison with the Shapley Policy in Zhang et al. (2024)). In Zhang et al. (2024), the authors study (3) with Wasserstein uncertainty sets, focusing on the newsvendor cost. They show that when optimization over all decision rules, the optimal decision rule, called Shapley policy, can be found by first solving for the in-sample Wasserstein robust optimal decision rule $\hat{f}_W$, and then extending to the entire space by solving

$$f_W(x) \in \arg \min_{w \in \mathbb{R}} \max_k \frac{|w - \hat{f}_W(\hat{x}_k)|}{\|x - \hat{x}_k\|}, \quad (\text{W}_{\text{Lip}})$$

Figure 5. (Color online) Intersection of $I_k(x)$ When $x = \mathbf{0}$ Is Empty



Note. $I_1(x)$ is the left region, $I_2(x)$ is the lower right region, and $I_3(x)$ is the upper right region.

which minimizes the maximal slope. Using the same idea, if we define

$$f_\infty(x) \in \arg \min_{w \in \mathbb{R}} \max_k \frac{|w - \hat{f}^*(\hat{x}_k)|}{\|x - \hat{x}_k\|}, \quad (\text{C}_{\text{Lip}})$$

where $\hat{f}^*(\hat{x}_k)$ s are defined in Theorem 3, then it can be verified that $f_\infty(x) \in \cap_k I_k(x)$. Therefore, this shows that $f_\infty(x)$ is a robust optimal decision rule for (11). Note that we use the subscript ∞ to indicate the ∞ -norm (maximum) of the slope function $k \mapsto |w - \hat{f}^*(\hat{x}_k)| / \|x - \hat{x}_k\|$.

Differently, we can define another decision rule that minimizes the 1-norm of the slope function:

$$f_1(x) \in \arg \min_{w \in \mathbb{R}} \sum_k \frac{|w - \hat{f}^*(\hat{x}_k)|}{\|x - \hat{x}_k\|}. \quad (\text{C}_{\text{TV}})$$

The resulting decision rule may not be optimal, but we can always truncate its values to ensure they fall within $\cap_k I_k(x)$ and thereby make it robustly optimal. Namely, if we use $\bar{I}(\cdot)$ and $\underline{I}(\cdot)$ to represent the upper and lower bound of the region $\cap_k I_k(x)$, respectively, then we define

$$\bar{f}_1(x) := \max(\underline{I}(x), \min(f_1(x), \bar{I}(x))). \quad (\text{C}_{\text{TV-trunc}})$$

We denote the truncated decision rule as $\bar{f}_1(x)$.

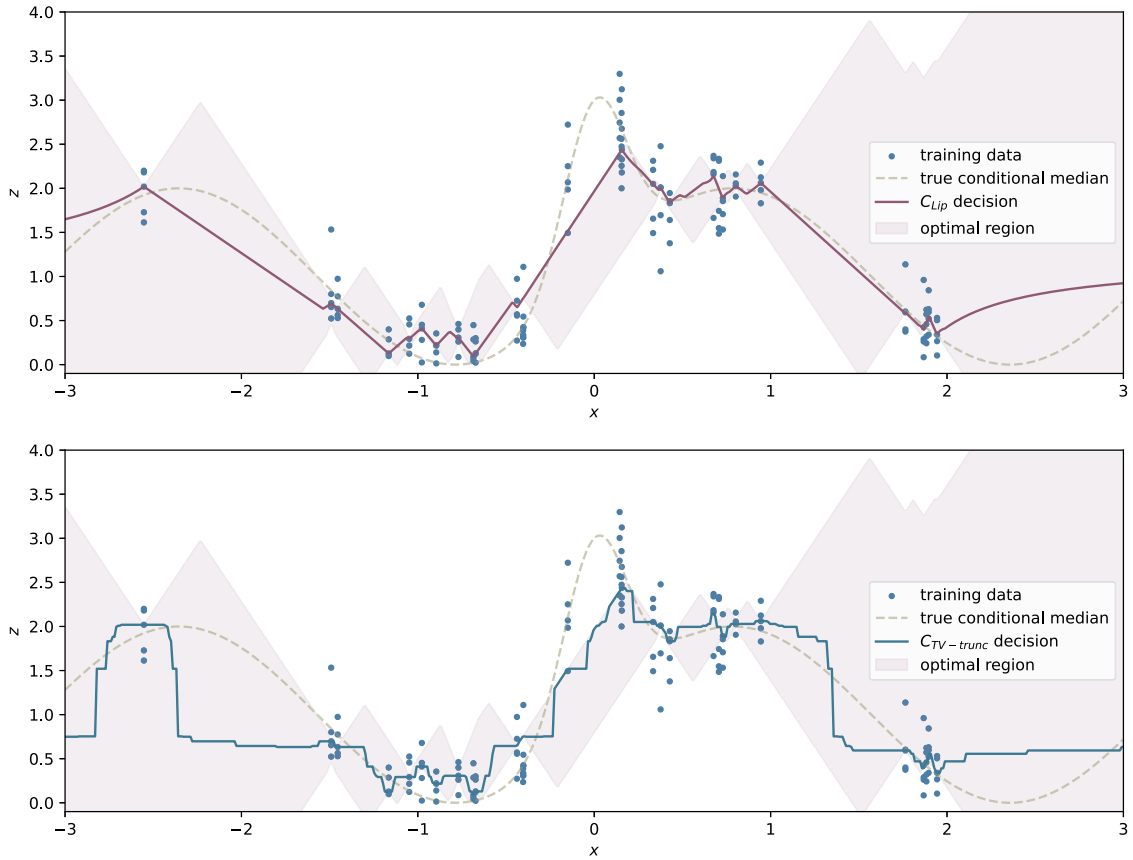
We illustrate the two robust optimal decision rules defined above using a conditional median estimate problem with $Z = \mu(X) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 1)$, $\mu(x) = \sin(2x) + 2\exp(-16x^2)$, as demonstrated in Figure 6. Based on the training data, one seeks to learn the conditional median expressed by the dashed line. After solving the in-sample causal problem, C_{Lip} and C_{TV} extend the same in-sample policy to different decision rules f_∞ and f_1 ; f_∞ is always in the optimal region, but f_1 needs to be clipped to fit in the optimal region.

Example 5 (Revisited). Consider the feature-based newsvendor problem in Example 5. When $h = b = 1$, this is equivalent to conditional median estimation. As detailed in Online Appendix EC.5, the in-sample dual problem (13) can be transformed into a linear programming problem:

$$\begin{aligned} \inf_{\{w_k\}_{k \in [K]}, \{y_k\}_{k \in [K]}, \{c_{kji}\}_{k,j,i \in [K]}, \lambda \geq 1} \quad & \lambda \rho + \sum_{k=1}^K y_k \\ \text{s.t.} \quad & y_j \geq \sum_{i=1}^{n_j} \hat{p}_{ki} (c_{kji} - \lambda \|\hat{x}_k - \hat{x}_j\|) \quad \forall j, k \in [K], \\ & c_{kji} \geq w_k - \hat{z}_{ji} \quad \forall k, j \in [K], i \in [n_j], \\ & c_{kji} \geq \hat{z}_{ji} - w_k \quad \forall k, j \in [K], i \in [n_j]. \end{aligned}$$

This is a linear programming with $K(n+2)+1$ variables and $K(2n+K)+1$ constraints.

Figure 6. (Color online) Two Robust Optimal Decision Rules f_∞ and $\bar{f}_1$ of a Median Estimation Problem



Notes. The horizontal axis represents x , and the vertical axis represents z . The dashed line represents the true conditional median $z = \mu(x)$.

Example 6 (Revisited). Consider the personalized pricing problem in Example 6. By Theorem 1, its strong dual problem can be written as

$$\inf_{\substack{f: \mathcal{X} \rightarrow \mathbb{R} \\ \lambda \geq 0}} \left\{ \lambda \rho^p + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\sup_{x \in \mathcal{X}} \left\{ \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|X}} \left[\sup_{z \in \mathcal{Z}} \left\{ -f(x) z^\top \begin{pmatrix} f(x) \\ 1 \end{pmatrix} \right. \right. \right. \right. \right. \\ \left. \left. \left. - \lambda \|z - \hat{Z}\|^p \right\} \right] \hat{X} \right] - \lambda \|x - \hat{X}\|^p \right\} \right\}.$$

In the case of $p = 1$, we notice that f is real-valued and Ψ is convex in w , so we may use Theorem 3 to reformulate the problem as

$$\inf_{\substack{\hat{f}: \hat{\mathcal{X}} \rightarrow \mathbb{R} \\ \lambda \geq 0}} \left\{ \lambda \rho + \mathbb{E}_{\hat{\mathbb{P}}_{\hat{X}}} \left[\max_{1 \leq k \leq K} \{ \varphi(\hat{f}(\hat{x}_k); \lambda, \hat{X}) - \lambda \|\hat{x}_k - \hat{X}\| \} \right] \right\}.$$

Here,

$$\varphi(w; \lambda; \hat{x}) = \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|X}} \left[\sup_{z \in \mathcal{Z}} \left\{ -w z^\top \begin{pmatrix} w \\ 1 \end{pmatrix} - \lambda \|z - \hat{Z}\| \right\} \mid \hat{X} = \hat{x} \right].$$

In particular, it can be reformulated as the following:

$$\begin{aligned} \inf_{\{w_k\}_k, \{c_k\}_k, \lambda \geq 0} \quad & \lambda \rho + \sum_{k \in [K]} \hat{p}_k c_k \\ \text{s.t.} \quad & c_j + (w_k^2 w_k) \bar{z}_k + \lambda \|\hat{x}_k - \hat{x}_j\| \geq 0 \quad \forall j, k \in [K], \\ & \|(w_k^2 w_k)\|_* \leq \lambda \quad \forall k \in [K]. \end{aligned} \quad (14)$$

Here, $\hat{p}_k = \sum_{i=1}^{n_k} \hat{p}_{ki}$ and $\bar{z}_k = \mathbb{E}_{\hat{\mathbb{P}}_{\hat{Z}|X}} [\hat{Z} \mid \hat{X} = \hat{x}_k]$; $\|\cdot\|_*$ is the dual norm of $\|\cdot\|_Z$. When $\mathcal{Z} = \mathbb{R}^2$ is equipped with

ℓ^1 or ℓ^∞ norm, (14) can be reduced to a quadratic program, whereas when the ℓ^2 norm is chosen, (14) can be written as a second-order conic program. A detailed calculation can be found in Online Appendix EC.5.

5. Numerical Experiments

5.1. Feature-Based Newsvendor

5.1.1. Synthetic Data. In this section, we illustrate our proposed approach in the context of a feature-based newsvendor. We consider a similar setup as in Zhang et al. (2024), where the demand Z depends on X in a nonlinear way:

$$Z = f(\beta^\top X) + \varepsilon, \quad f(\lambda) := c[\sin(2\lambda) + 2\exp(-16\lambda^2) + 1],$$

where $\varepsilon \sim \mathcal{N}(0, 1)$ is a standard Gaussian variable independent from β and X . Let the coefficient vector $\beta \in \mathbb{R}^{100}$, with each component independently sampled from a uniform distribution $\mathcal{U}([-0.1, 0.1])$. The covariate X is sampled from a 100-dimensional multivariate normal distribution $\mathcal{N}(0, (\sigma_{ij})_{ij})$, with mean zero and covariate matrix defined by $\sigma_{ij} = 0.5^{|i-j|}$ with $i, j = 1, \dots, 100$. The constant $c = 1.7$ is chosen such that the signal-to-noise ratio is approximately 3:1. Because the demand should be positive, we reject all samples with $Z < 0$.

We experiment with different unit overage cost $h \in \{0.2, 0.5, 0.8, 1\}$ while fixing the unit underage cost $b = 1$. To understand the effect of the sample size, we choose $K \in \{10, 30, 100, 300\}$ and $n_k \in \{1, 3, 10, 30, 100\}$. The testing data size is 10,000. The hyperparameters are tuned based on fivefold cross-validation. We set $\|\cdot\|_{\mathcal{X}} = \|\cdot\|_2$ and $\|\cdot\|_{\mathcal{Z}} = \infty \cdot \mathbf{1}\{z \neq \hat{z}\}$. To generate the boxplots, we run 20 repeated experiments (except for $K = 10$, we run 50 experiments to get a more accurate depiction). All experiments are performed in Ubuntu 18.04 using Python 3.6.9 with a convex optimization solver Gurobi 9.1.1, on a Dell Precision 5820 Tower Workstation with Intel Xeon W-2125 CPU (32 cores) and 32 GB RAM (DDR4 2666 MHz). Because of constraints associated with the solver's capabilities, the experiments with $n_k = 30$, $K = 300$ and $n_k = 100$, $K = 100, 300$ are not included in the comparison.

In our first set of experiments, we delve into the effects of adopting different distributional uncertainty sets of the inner worst-case expectation, namely, the Wasserstein DRO with Shapley extension W_{Lip} in Zhang et al. (2024) versus causal transport DRO with Shapley extension (C_{Lip}). Both approaches incorporate the Shapley extension to extend the in-sample optimal policy. Figure 7 shows the relative difference in the out-

of-sample expected cost between C_{Lip} and W_{Lip} with the same training and testing data set—a negative number indicates that C_{Lip} outperforms W_{Lip} .

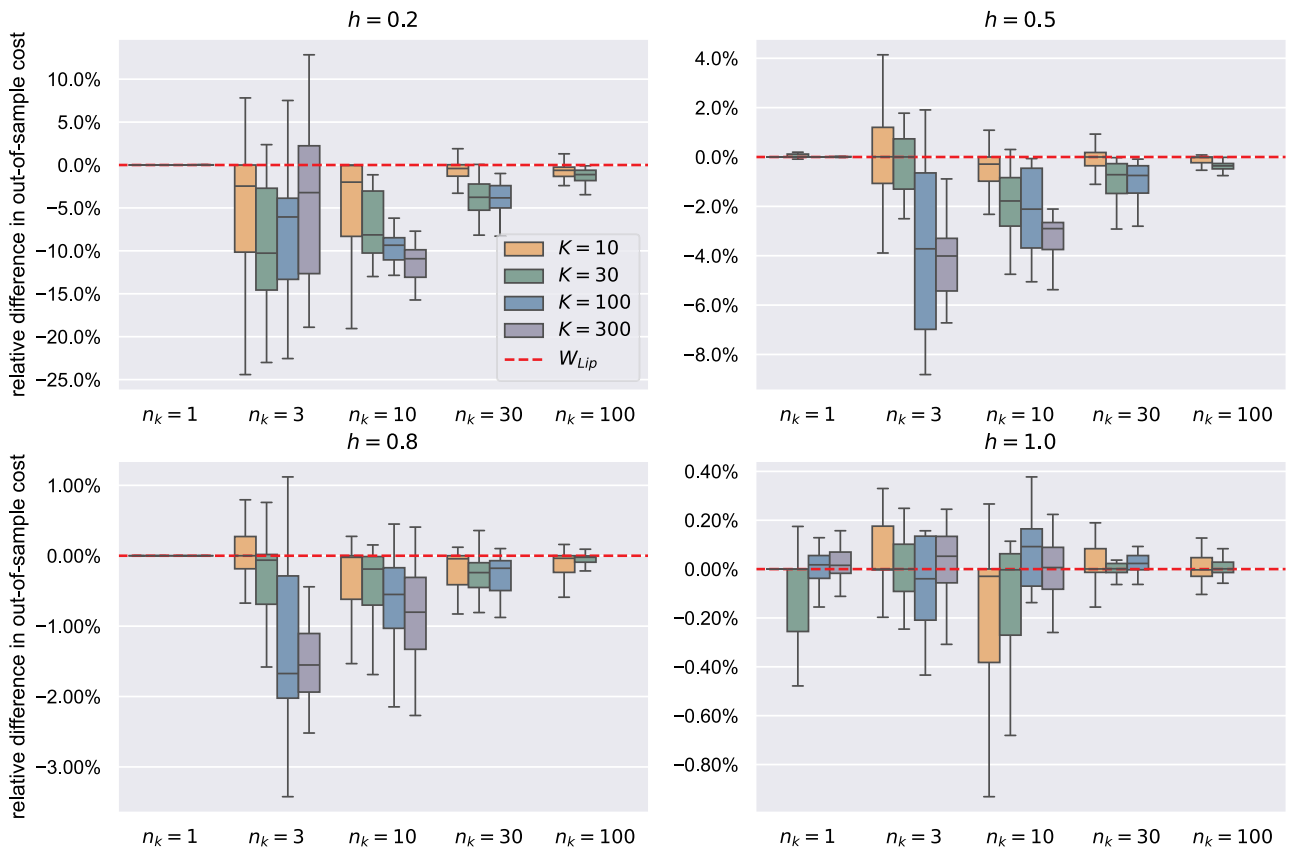
We have the following observations.

(I) When each covariate group contains only a single sample ($n_k = 1$), C_{Lip} and W_{Lip} have the same performance because the two formulations are equivalent (Remark 1).

(II) As the sample size per covariate group increases beyond a single sample, C_{Lip} begins to exhibit a performance advantage over W_{Lip} , particularly when dealing with skewed loss functions ($h = 0.2, 0.5, 0.8$). This edge is most pronounced at lower sample sizes $n_k = 3, 10$, which shows the value of (even a little) conditional information. The marginal benefit provided by C_{Lip} tends to diminish with larger sample sizes per covariate group ($n_k = 30, 100$). One explanation is that the worst-case distribution of W_{Lip} does not significantly degrade the conditional information structure when there are many samples at the same covariate value.

(III) The comparative advantage of C_{Lip} over W_{Lip} generally amplifies with the increase in the number of covariate groups K . An explanation is that when K is large, C_{Lip} can fully take advantage of the conditional information to extrapolate other conditional distributions.

Figure 7. (Color online) Boxplots of the Relative Differences in the Out-of-Sample Performance Between W_{Lip} (Baseline) and C_{Lip}



In addition to our main comparisons above, we further investigate two supplementary scenarios. The first one compares the Shapley extension versus non-Shapley extensions in the nonparametric setting, as discussed in Theorem 3 and Remark 3. The second compares truncated versus nontruncated causal transport DRO solutions, as identified in Theorem 3. We refer the two additional experiments in Online Appendix EC.6.

5.1.2. Real Data. We use a real-world data set from a meal delivery company (Genpact 2019, Qi et al. 2024), which records 145 weeks of historical weekly demand. The company fulfills customer meal orders using perishable ingredients, making its weekly procurement planning naturally fit into a newsvendor framework. In addition to demand data, the data set includes contextual features such as checkout price, promotion email indicators, and whether the meal was featured on the company’s homepage. We compare the empirical performance of our proposed C_{Lip} against several benchmarks, including empirical risk minimization using affine policy with ℓ^1 and ℓ^2 regularization ($\text{ERM2}(\ell^1/\ell^2)$) and kernel-weights optimization (KO) in Ban and Rudin (2019); conditional stochastic optimization using random forests (RandForest) and k -nearest neighbors (k NN) in Bertsimas and Kallus (2020); and stochastic optimization forests with different splitting criteria (StochOptForest (apx-soln/apx-risk)) in Kallus and Mao (2023).

We train all models on the most recent one year (52 weeks) of data using a rolling window and then test them out-of-sample on subsequent periods with planning horizons of 1, 4, and 12 weeks—corresponding to weekly, monthly, and quarterly reoptimization. In

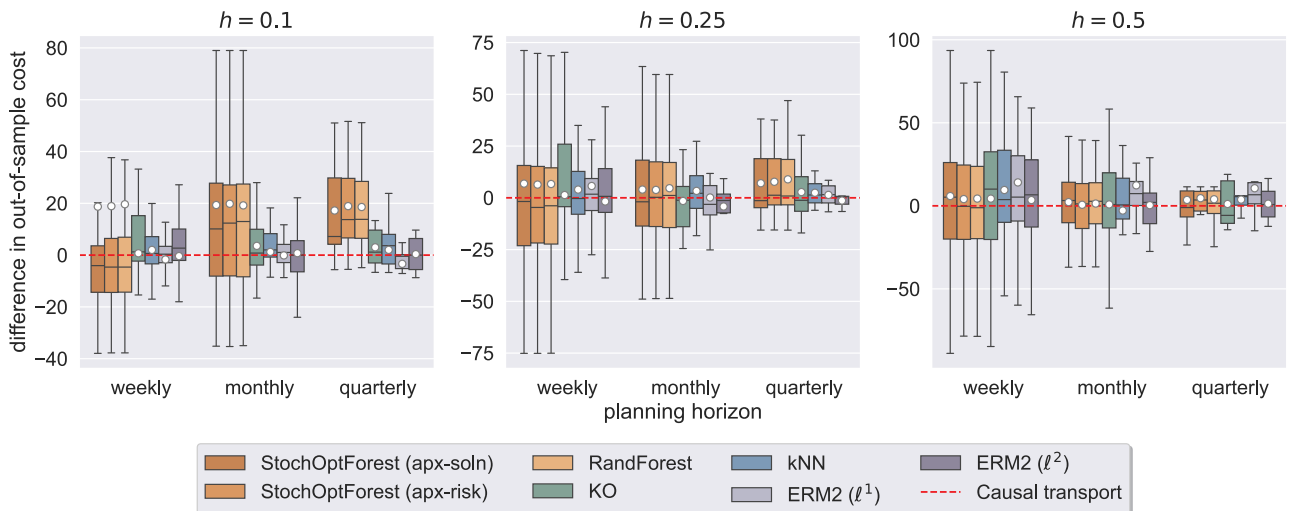
Figure 8, we compare differences in testing cost across competing methods under three levels of $h = 0.1, 0.25, 0.5$ —which are chosen to reflect the practical scenarios—and three planning horizons. The y -axis represents the cost difference relative to our C_{Lip} (dashed line at zero), with positive values indicating higher cost (worse performance) compared with C_{Lip} .

From Figure 8, we see that C_{Lip} achieves consistently strong performance across all nine configurations (three h values $\times$ three planning horizons), both in terms of mean performance (white circle) and median performance (black line). For small h , $\text{ERM2}(\ell^1/\ell^2)$ sometimes attains better performance, although without consistency. In the weekly planning horizon, forest-based methods (StochOptForest (apx-soln/apx-risk), RandForest) sometimes obtain lower median costs than C_{Lip} at $h = 0.1$ and 0.25 , but this comes with much higher variability and worse mean performance. Overall, the results indicate that C_{Lip} consistently delivers strong, stable out-of-sample performance.

5.2. Contextual Linear Optimization

This section presents numerical experiments on the contextual linear optimization problem from Example 7, examined in the setting of portfolio optimization. The goal is to determine the optimal weights $w \in \mathbb{R}^m$ across m assets in order to maximize the portfolio return $w^\top Z$, where $Z \in \mathbb{R}^m$ denotes the random return. To align with the setting in (1), we recast it as a minimization problem by defining the loss function $\Psi(w, z) = -w^\top z$, that is, the negative of the portfolio return. The allocation weights are subject to three constraints. First, the budget constraint requires $w^\top \mathbf{1} = 1$. Second, short-selling is permitted but limited, in the sense that no individual position may be shorter than

Figure 8. (Color online) Boxplots of the Relative Differences in the Out-of-Sample Performance Between C_{Lip} and Other Benchmarks



30%. Finally, the $m = 10$ assets are grouped into three sectors, $\{1, 2, 3\}$, $\{4, 5, 6, 7\}$, $\{8, 9, 10\}$, and the average weight within each group must be nonnegative, preventing systematic shorting of an entire sector.

The contexts are uniformly sampled from the ellipsoid $\mathcal{X} = \{x \in \mathbb{R}^d : x^\top \Sigma x \leq 1\}$, with $\Sigma = (\sigma_{ij})$ where $\sigma_{ij} = 0.5^{|i-j|}$ with $i, j = 1, \dots, d$, where $d = 5$. Given a context X , the random variable $Z \in \mathbb{R}^m$ is generated according to the following nonlinear model inspired by Feng et al. (2024):

$$Z_i = \sum_{j=1}^d f^{(\beta_{ij})}(X_j) + \sigma \left(\sum_{j=1}^d f^{(4)}(X_j) + 1 \right) \cdot \varepsilon_i, \quad i \in [m].$$

Here, the nonlinear functions are defined as

$$f^{(1)}(x) = \sin(2\pi x), f^{(2)}(x) = 0.5 \exp x, \\ f^{(3)}(x) = 1.5|(x - 0.4)(x - 0.6)|, f^{(4)}(x) = x \cos(2\pi x).$$

The indices β_{ij} are independently sampled from $\{1, 2, 3\}$ at the beginning of each experiment and remain unchanged within that experiment. The noise scaling factor $\sigma \in \{1, 3, 10\}$ controls the signal-to-noise ratio. The noise ε_i follows a t -distribution with three degrees of freedom. We generate K context samples of X . For each sample $\hat{x}_k$, we then generate n_k samples of Z , resulting $n = \sum_{k \in [K]} n_k$ data pairs $(\hat{x}_k, \hat{z}_{ki})$.

To further investigate the comparison between Wasserstein and causal transport uncertainty sets with affine policy class, we vary the number of context samples $K \in \{20, 50, 100\}$, the number of outcome samples per context $n_k \in \{10, 30, 100\}$, and the noise level $\sigma \in \{1, 3, 10\}$. Figure 9 reports boxplots of the out-of-sample cost of Wasserstein DRO policy (W-DRO) and Causal DRO policy (C-DRO) compared with an oracle model using ERM1 algorithm introduced in Ban and

Rudin (2019) on a training data set of 10^7 samples. Across nearly all settings, C-DRO outperforms W-DRO in terms of median cost, with the performance gap widening as the noise level increases. At a low noise level, the gap narrows as the number of contexts increases. Consistent with the newsvendor experiments in Section 5.1.1, when a moderate amount of conditional information is available, C-DRO achieves a clear advantage over W-DRO. In contrast, when conditional information is either scarce or overly abundant, the difference between the two methods diminishes. This is likely because, in the former case, C-DRO cannot exploit the limited conditional information, whereas in the latter case, the worst-case distribution of W-DRO does not significantly distort the underlying conditional structure.

We also investigate the impact of the distribution shift. Fixing $\sigma = 1$, $n_k = 30$, we use the same data above to train the model, but the testing data distribution has a different conditional distribution $Z|X$ (known as concept shift):

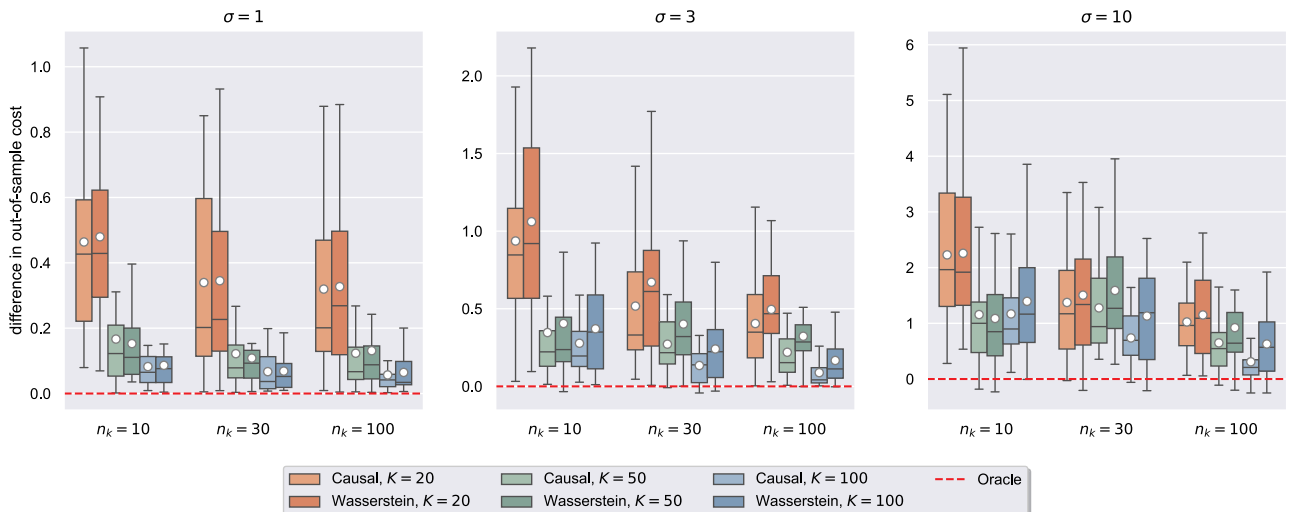
$$Z_i^{\text{test}} = \sum_{j=1}^d f^{(\beta_{ij})}(X_j) + \sigma \left(\sum_{j=1}^d f^{(4)}(X_j) + 1 \right) \cdot \varepsilon_i + \sigma_{\text{shift}} \cdot \varepsilon_i^{\text{shift}}, \\ i \in [m].$$

Here, $\sigma_{\text{shift}} \in \{1, 3, 10\}$ is a scaling factor of the distributional shift, and $\varepsilon_i^{\text{shift}} \sim \mathcal{N}(\mu_i^{\text{shift}}, \sigma_i^{\text{shift}})$ are independent normal random variables, with context-dependent mean and variance given by

$$\mu_i^{\text{shift}} = \sum_{j=1}^d f^{(\beta_{ij}^{\text{mean}})}(X_j), \quad \sigma_i^{\text{shift}} = \sum_{j=1}^d f^{(\beta_{ij}^{\text{var}})}(X_j),$$

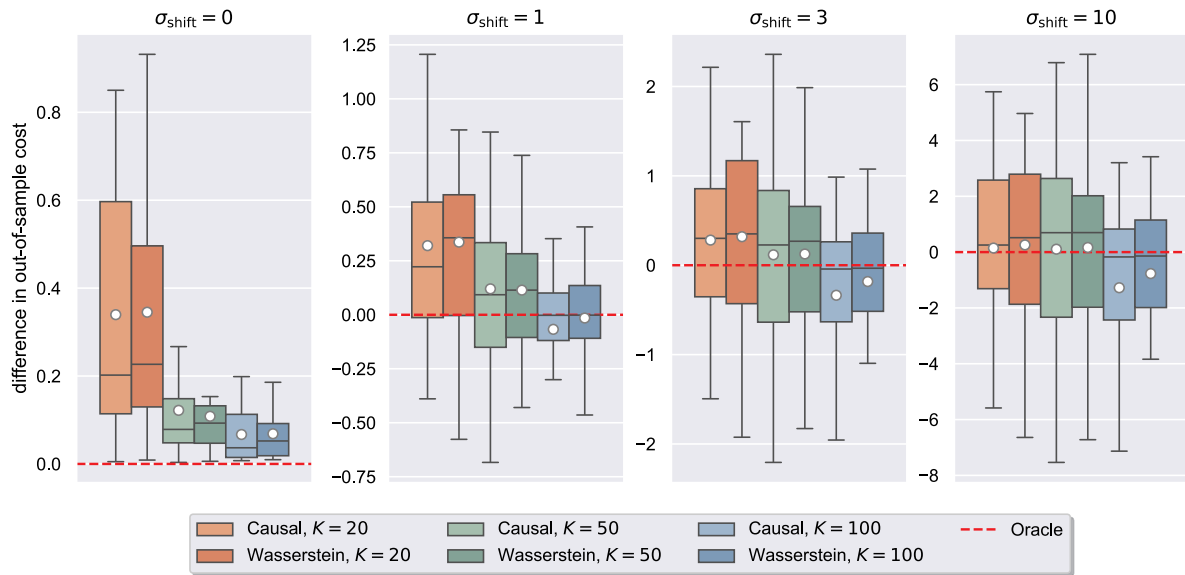
where β_{ij}^{mean} and β_{ij}^{var} are uniformly sampled from

Figure 9. (Color online) Experiment with Fixed Number n_k of z per x Chosen in $\{10, 30, 100\}$, with Noise Level $\sigma \in \{1, 3, 10\}$ and Number of Contexts $K \in \{20, 50, 100\}$



Note. They are compared with a baseline scenario that applies the ERM1 algorithm on a much larger training data set.

Figure 10. (Color online) Experiment with Concept Shift, with Shift Level $\sigma_{\text{shift}} \in \{0, 1, 3, 10\}$ and Number of Contexts $K \in \{20, 50, 100\}$



$\{1, 2, 3\}$, independently from β_{ij} and each other, but are fixed within each experiment.

Figure 10 reports boxplots of the out-of-sample cost of W-DRO and C-DRO compared again with the baseline using ERM1 algorithm with 10^7 samples. We observe that C-DRO consistently outperforms W-DRO, especially with a large concept shift σ_{shift} and a large number of contexts K . This demonstrates the improved robustness of C-DRO in hedging against shifts in the conditional distributions.

6. Concluding Remarks

In this paper, we propose a new distributionally robust decision rule optimization for decision making with side information based on the causal transport distance. These results open up new research directions for distributionally robust optimization and adjustable robust optimization. For future work, it would be interesting to investigate the performance guarantees of the proposed framework.

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